

| <b>VII Semester</b>                                                                           |                    |                                                                        |                     |
|-----------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------|---------------------|
| <b>S. No</b>                                                                                  | <b>Course Code</b> | <b>Course Title</b>                                                    | <b>Credit Hours</b> |
| 1.                                                                                            | <b>AECON 411</b>   | Agri-business Management**                                             | 4(3+1)              |
| 2.                                                                                            | <b>AGRON-411</b>   | Management of Natural Resources**                                      | 4(3+1)              |
| 3.                                                                                            | <b>AGRON-412</b>   | System Simulation and Agro-advisory**                                  | 4(3+1)              |
| 4.                                                                                            | <b>AGRON-413</b>   | Climate Resilient Agriculture**                                        | 4(3+1)              |
| 5.                                                                                            | <b>AGRON-414</b>   | Geo-informatics and Remote Sensing, Precision Farming**                | 4(3+1)              |
| 6.                                                                                            | <b>AGRON-415</b>   | Principles and Practices of organic Farming/Conservation Agriculture** | 4(3+1)              |
| 7.                                                                                            | <b>AENGG-411</b>   | Protected Cultivation**                                                | 4(3+1)              |
| 8.                                                                                            | <b>AGEXT-411</b>   | Agricultural Journalism**                                              | 4(3+1)              |
| 9.                                                                                            | <b>BT-411</b>      | Biotechnology of Crop Improvement**                                    | 4(3+1)              |
| 10.                                                                                           | <b>ENTO-411</b>    | Bio-pesticides & Bio-fertilizers**                                     | 4(3+1)              |
| 11.                                                                                           | <b>ENTO-412</b>    | Bio-formulation and Nano Formulation**                                 | 4(3+1)              |
| 12.                                                                                           | <b>FSN-411</b>     | Food Safety and Standards**                                            | 4(3+1)              |
| 13.                                                                                           | <b>FSN-412</b>     | Food Science & Nutrition**                                             | 4(3+1)              |
| 14.                                                                                           | <b>GPB-411</b>     | Commercial Plant Breeding**                                            | 4(3+1)              |
| 15.                                                                                           | <b>GPB-412</b>     | Micro-propagation Technologies**                                       | 4(3+1)              |
| 16.                                                                                           | <b>GPB-413</b>     | Commercial Seed Production**                                           | 4(3+1)              |
| 17.                                                                                           | <b>HORT-411</b>    | Landscaping**                                                          | 4(3+1)              |
| 18.                                                                                           | <b>HORT-412</b>    | Hi-tech Horticulture**                                                 | 4(3+1)              |
| 19.                                                                                           | <b>HORT-413</b>    | Post-harvest Technology and Value Addition**                           | 4(3+1)              |
| 20.                                                                                           | <b>SSAC-411</b>    | Vermicompost and Vermiculture**                                        | 4(3+1)              |
| <b>5 Elective Course (Major or Minor) will be selected compulsory each of 4 (3+1) credits</b> |                    |                                                                        |                     |
| <b>Total Credit</b>                                                                           |                    |                                                                        | <b>20 (15+5)</b>    |

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| <b>AECON 411</b> | <b>Agri-business Management**</b> | <b>4(3+1)</b> |
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### Objectives

1. To impart knowledge on understanding the concepts processes, significance, and role of management and organizational behavior
2. To impart knowledge on external and internal environment of a business and its management

### Theory

Transformation of agriculture into agribusiness, various stakeholders and components of agribusiness systems. Importance of agribusiness in the Indian economy and New Agricultural Policy. Distinctive features of Agribusiness Management: Importance and needs of agro-based industries. Classification of industries and types of agro based industries. Institutional arrangement, procedures to set up agro based industries. Constraints in establishing agro-based industries. Agri-value chain: Understanding primary and support activities and their linkages. Business environment: PEST and SWOT analysis. Management functions: Roles and activities, Organization culture. Planning, meaning, definition, types of plans. Purpose or mission, goals or objectives, Strategies, policies procedures, rules, programs and budget. Components of a business plan, Steps in planning and implementation. Organization staffing, directing and motivation. Ordering, leading, supervision, communications, control. Capital management and Financial management of Agribusiness. Financial statements and their importance. Marketing Management: Segmentation, targeting and positioning. Marketing mix and marketing strategies. Consumer behaviour analysis, Product Life Cycle (PLC). Sales and Distribution Management. Pricing policy, various pricing methods. Project Management definition, project cycle, identification, formulation, appraisal, implementation, monitoring and evaluation. Project Appraisal and evaluation techniques.

### Practical

Study of agri –input markets: Seed, fertilizers pesticides. Study of output markets: grains, fruits, vegetables, flowers. Study of product market, retails trade commodity trading, and value-added products. Study of financing institutions- Cooperative, Commercial Bank, RRBs, Agribusiness Finance Limited, NABARD. Preparations of projects and Feasibility reports for agri business entrepreneur. Appraisal /evaluation techniques for identifying viable projects- Non-discounting techniques. Case study of agro- based industries. Trend and growth rate of price of agricultural commodities. Net present worth technique for selection of viable project. Internal rate of return.

### Lecture Schedule: Theory

| <b>S. No.</b> | <b>Name of Topic</b>                                          | <b>No. of Lectures</b> |
|---------------|---------------------------------------------------------------|------------------------|
| 1.            | Transformation of agriculture into agribusiness: Introduction | 1                      |
| 2.            | Various stakeholders and components of agribusiness systems   | 2                      |

|     |                                                                                                           |   |
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| 3.  | Importance of agribusiness in the Indian economy and New Agricultural Policy                              | 2 |
| 4.  | Distinctive features of Agribusiness Management: Importance and needs of agro-based industries            | 2 |
| 5.  | Classification of industries and types of agro-based industries                                           | 2 |
| 6.  | Institutional arrangement and procedures to set up agro-based industries                                  | 1 |
| 7.  | Constraints in establishing agro-based industries                                                         | 1 |
| 8.  | Agri-value chain: Understanding primary and support activities and their linkages                         | 3 |
| 9.  | Business environment: Introduction to PEST analysis and application                                       | 1 |
| 10. | Business environment: SWOT analysis                                                                       | 1 |
| 11. | Management functions: Roles and activities overview                                                       | 1 |
| 12. | Organization culture: Importance and impact                                                               | 1 |
| 13. | Planning: Meaning and definition, Types of plans: Short-term and long-term plans,                         | 2 |
| 14. | Purpose or mission, goals or objectives Strategies, policies, procedures, rules, programs, and budgets    | 2 |
| 15. | Components of a business plan, Steps in planning and implementation                                       | 1 |
| 16. | Organization staffing, directing and motivation, Ordering, leading, supervision, communications, control. | 2 |
| 17. | Capital management and financial management in agribusiness                                               | 1 |
| 18. | Financial statements and their importance                                                                 | 2 |
| 19. | Marketing management: Segmentation, targeting, and positioning                                            | 2 |
| 20. | Marketing mix: Concept and components                                                                     | 1 |
| 21. | Marketing strategies: Overview and importance                                                             | 1 |
| 22. | Consumer behaviour analysis: Factors and models                                                           | 1 |
| 23. | Product Life Cycle (PLC): Stages and implications                                                         | 1 |
| 24. | Sales and distribution management in agribusiness                                                         | 1 |
| 25. | Pricing policy: Overview                                                                                  | 1 |
| 26. | Various pricing methods: Analysis and application                                                         | 1 |
| 27. | Project management: Definition and importance                                                             | 1 |
| 28. | Project cycle: Stages and overview                                                                        | 1 |
| 29. | Project Appraisal and evaluation techniques.                                                              | 3 |

**Lecture Schedule: Practical**

| <b>S. No.</b> | <b>Name of Topic</b>                                                                  | <b>No. of Lectures</b> |
|---------------|---------------------------------------------------------------------------------------|------------------------|
| 1.            | Study of agri-input markets: Overview of seed market                                  | 1                      |
| 2.            | Study of agri-input markets: Fertilizers and Pesticides                               | 1                      |
| 3.            | Study of output markets: Grains market analysis                                       | 1                      |
| 4.            | Study of output markets: Fruits and vegetables markets                                | 1                      |
| 5.            | Study of output markets: Flower markets and their potential                           | 1                      |
| 6.            | Study of product market, retails trade, Commodity trading: Mechanisms and platforms   | 1                      |
| 7.            | Value-added products: Market opportunities and challenges                             | 1                      |
| 8.            | Study of financing institutions: Cooperative structures                               | 1                      |
| 9.            | Study of financing institutions: Commercial Banks and RRBs                            | 1                      |
| 10.           | Agribusiness Finance Limited and its role                                             | 1                      |
| 11.           | NABARD: Structure, function, and support for agribusiness                             | 1                      |
| 12.           | Preparation of agribusiness projects: Initial steps                                   | 1                      |
| 13.           | Feasibility reports: Components and preparation techniques                            | 1                      |
| 14.           | Appraisal and evaluation techniques: Introduction to project analysis                 | 3                      |
| 15.           | Non-discounting techniques for project appraisal: Methods and application             | 1                      |
| 16.           | Case study of successful agro-based industries: Analysis Part 1                       | 1                      |
| 17.           | Case study of successful agro-based industries: Analysis Part 2                       | 1                      |
| 18.           | Trend and growth rate of price of agricultural commodities                            | 1                      |
| 19.           | Net present worth technique for selection of viable project. Internal rate of return. | 1                      |

**Suggested readings**

1. Broadway, A. C. and Broadway, Arif, A. 2002. A text book of Agri-Business Management. Kalyani Publishers
2. Bairwa, S. L. 2016. Objective on Fundamentals of Agri-business Management. Kalyani Publishers
3. Anjan Nishra, Debasish Biswas and Arunangshu Giri. 2019. Agribusiness Management, Himalaya Publishing House, 220 p.
4. Bairwa, S. L., Chandra Sen, L. K. Meena and Meera Kumari. 2018. Agribusiness Management Theory and Practices, Write and Print Publications.

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| <b>AGRON-411</b> | <b>Management of Natural Resources**</b> | <b>4(3+1)</b> |
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### **Objectives**

1. To enlighten students about available natural resources and their relationship with crop Production
2. To impart the knowledge of principles and practices of natural resource management

### **Theory**

Introduction to Natural Resource Bases: Concept of resource, classification of natural resources. Factors influencing resource availability, distribution and uses. Interrelationships among different types of natural resources. Concern on Productivity issues. Ecological, social and economic dimension of resource management. Land resources: Land as a resource. Dry land, land use classification, land degradation, man induced landslides, soil erosion and desertification. Landscape impact analysis, wetland ecology and management. Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. Water ecology and management. Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources. Resource Management Paradigms: Resource management the evolution and history of resource management paradigms. Resource conflicts: Resource extraction, access and control system. Approaches in Resource Management: Ecological approach; economic approach; ethnological approach; implications of the approaches; integrated resource management strategies. Introduction to soil and water conservation and causes of soil erosion., Definition and agents of soil erosion, water erosion - Forms of water erosion, Gully classification and control measures. Soil loss estimation by universal soil loss equation - Soil loss measurement techniques. Principles of erosion control - Introduction to contouring, strip cropping. Contour bund - Graded bund and bench terracing. Wind erosion - Mechanics of wind erosion, types of soil movement - Principles of wind erosion control and its control measures, Water harvesting techniques - Lining of ponds, tanks and canal systems.

### **Practical**

Identifying natural resources and their utility. Practicing survey - Principles and educating to use pacing technique for measurement. Area calculations through chain survey - GPS demo for tracking and area measurement. Estimation of soil loss and calculation of erosion index. Leveling concepts and practical utility in agriculture. Preparation of contour maps. Concept of vegetative water ways and design of grassed water ways. Wind erosion and estimation process. Different irrigation pumps and their constructional differences. Farm pond construction and its

design aspects. Visit to nearby farm pond. Visit to an erosion site. Exposure to strip cropping/contour bunding.

**Lecture schedule: Theory**

| S.N. | Topic                                                                                                                                     | No. of lectures |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Introduction to Natural Resource Bases: Concept of resource, classification of natural resources                                          | 2               |
| 2.   | Factors influencing resource availability, distribution and uses                                                                          | 2               |
| 3.   | Interrelationships among different types of natural resources                                                                             | 2               |
| 4.   | Concern on Productivity issues                                                                                                            | 2               |
| 5.   | Ecological, social and economic dimension of resource management                                                                          | 2               |
| 6.   | Land resources: Land as a resource                                                                                                        | 2               |
| 7.   | Dry land, land use classification, land degradation, man induced landslides, soil erosion and desertification                             | 2               |
| 8.   | Landscape impact analysis, wetland ecology and management.                                                                                | 2               |
| 9.   | Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. | 3               |
| 10.  | Water ecology and management                                                                                                              | 2               |
| 11.  | Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources                        | 3               |
| 12.  | Resource Management Paradigms: Resource management the evolution and history of resource management paradigms                             | 2               |
| 13.  | Resource conflicts: Resource extraction, access and control system                                                                        | 2               |
| 14.  | Approaches in Resource Management: Ecological approach; economic approach                                                                 | 2               |
| 15.  | Ethnological approach; implications of the approaches; integrated resource management strategies                                          | 2               |
| 16.  | Introduction to soil and water conservation and causes of soil erosion., Definition and agents of soil erosion                            | 2               |
| 17.  | Water erosion - Forms of water erosion, Gully classification and control measures                                                         | 2               |
| 18.  | Soil loss estimation by universal soil loss equation - Soil loss measurement techniques                                                   | 2               |
| 19.  | Principles of erosion control - Introduction to contouring, strip cropping.                                                               | 2               |
| 20   | Contour bund - Graded bund and bench terracing                                                                                            | 2               |

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|-----|-------------------------------------------------------------------------|---|
| 21. | Wind erosion - Mechanics of wind erosion, types of soil movement        | 2 |
| 22. | Principles of wind erosion control and its control measures             | 2 |
| 23. | Water harvesting techniques - Lining of ponds, tanks and canal systems. | 2 |

### Lecture schedule: Practical

| S.N. | Topic                                                                                | No. of lectures |
|------|--------------------------------------------------------------------------------------|-----------------|
| 1.   | Identifying natural resources and their utility.                                     | 1               |
| 2.   | Practicing survey - Principles and educating to use pacing technique for measurement | 2               |
| 3.   | Area calculations through chain survey - GPS demo for tracking and area measurement  | 2               |
| 4.   | Estimation of soil loss and calculation of erosion index                             | 1               |
| 5.   | Leveling concepts and practical utility in agriculture                               | 1               |
| 6.   | Preparation of contour maps                                                          | 1               |
| 7.   | Concept of vegetative water ways and design of grassed water ways.                   | 1               |
| 8.   | Wind erosion and estimation process                                                  | 1               |
| 9.   | Different irrigation pumps and their constructional differences                      | 2               |
| 10.  | Farm pond construction and its design aspects                                        | 1               |
| 11.  | Visit to nearby farm pond                                                            | 1               |
| 12.  | Visit to an erosion site                                                             | 1               |
| 13.  | Exposure to strip cropping/contour bunding                                           | 1               |

### Suggested readings

1. Sustainable Natural Resource Management by Danill R. Lynch.
2. Management of Natural Resource for Sustainable Development, by Vijay Singh Rathor and B S Rathor, Daya Publishing House.
3. Managing Natural Resources: Focus on Land and Water. Ed. Harikesh N. Mishra. PHI, Learning, 496p.
4. Management of Resources for Sustainable Development: Sushma Goel. The Orient Blackswan 284p.
5. Natural Resources: Their Conservation and Management by Arvindrai Upadhyay. Aspiration Academy, 320p.
6. Natural Resource Management for Growth Development and Sustainability by Vasudeva Srishti Pal. Today & Tomorrows Printers and Publishers, 336p.
- 7.

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|------------------|----------------------------------------------|---------------|
| <b>AGRON-412</b> | <b>System Simulation and Agro-advisory**</b> | <b>4(3+1)</b> |
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### Objectives

1. To impart the knowledge of statistical and simulation modelling in crop yield estimation

2. To get acquainted with different weather forecasting techniques and their usability analysis
3. To study about the preparation and dissemination of agro-advisory bulletin

### **Theory**

System Approach for representing soil-plant-atmospheric continuum, system boundaries, Crop models, concepts & techniques, types of crop models, data requirements, relational diagrams. Evaluation of crop responses to weather elements; Elementary crop growth models; calibration, validation, verification and sensitivity analysis. Potential and achievable crop production- concept and modelling techniques for their estimation. Crop production in moisture and nutrients limited conditions; components of soil water and nutrients balance. Weather forecasting, types, methods, tools & techniques, forecast verification; Value added weather forecast, ITK for weather forecast and its validity; Crop-Weather Calendars; Preparation of agro-advisory bulletin based on weather forecast. Use of crop simulation model for preparation of Agro-advisory and its effective dissemination.

### **Practical**

Preparation of crop weather calendars. Preparation of agro-advisories based on weather forecast using various approaches and synoptic charts. Working with statistical and simulation models for crop growth. Potential & achievable production; yield forecasting, insect & disease forecasting models. Simulation with limitations of water and nutrient management options. Sensitivity analysis of varying weather and crop management practices. Use of statistical approaches in data analysis and preparation of historical, past and present meteorological data for medium range weather forecast. Feedback from farmers about the agro-advisory.

### **Lecture schedule: Theory**

| S.N. | Topic                                                                                          | No. of lectures |
|------|------------------------------------------------------------------------------------------------|-----------------|
| 1.   | System Approach for representing soil-plant-atmospheric continuum                              | 2               |
| 2.   | System boundaries, Crop models, concepts & techniques                                          | 2               |
| 3.   | Types of crop models, data requirements, relational diagrams                                   | 3               |
| 4.   | Evaluation of crop responses to weather elements                                               | 2               |
| 5.   | Elementary crop growth models; calibration, validation, verification and sensitivity analysis  | 3               |
| 6.   | Potential and achievable crop production- concept and modeling techniques for their estimation | 2               |



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| 7.  | Crop production in moisture and nutrients limited conditions; components of soil water and nutrients balance | 3 |
| 8.  | Weather forecasting, types, methods, tools & techniques                                                      | 3 |
| 9.  | forecast verification; Value added weather forecast                                                          | 2 |
| 10  | ITK for weather forecast and its validity                                                                    | 2 |
| 11. | Crop-Weather Calendars                                                                                       | 2 |
| 12. | Preparation of agro-advisory bulletin based on weather forecast                                              | 3 |
| 13  | Use of crop simulation model for preparation of Agro-advisory and its effective dissemination                | 3 |

### Lecture schedule: Practical

| S.N. | Topic                                                                                                                                                | No. of lectures |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Preparation of crop weather calendars                                                                                                                | 1               |
| 2.   | Preparation of agro-advisories based on weather forecast using various approaches and synoptic charts                                                | 2               |
| 3.   | Working with statistical and simulation models for crop growth                                                                                       | 2               |
| 4.   | Potential & achievable production; yield forecasting                                                                                                 | 2               |
| 5.   | insect & disease forecasting models                                                                                                                  | 1               |
| 6.   | Simulation with limitations of water and nutrient management options                                                                                 | 2               |
| 7.   | Sensitivity analysis of varying weather and crop management practices                                                                                | 1               |
| 8.   | Use of statistical approaches in data analysis and preparation of historical, past and present meteorological data for medium range weather forecast | 3               |
| 9.   | Feedback from farmers about the agro-advisory                                                                                                        | 2               |

### Suggested readings

1. Introduction to Agrometeorology by H. S. Mavi.
2. Agricultural Meteorology by G.S.L.H.V. Prasado Rao.
3. Advances in Plant Atmospheric Interactions (Eds. Rao, V.U.M., Rao, A.V.M.S., Rao, G.G.S.N., Ramana Rao, B.V., Vijaya Kumar, P. and Venkateswarlu, B), Central Research Institute for Dryland Agriculture (CRIDA), Santoshnagar, Hyderabad.
4. Text Book of Agricultural Meteorology by M.C. Varshneya and P.B. Pillai. ICAR.
5. Principles of Agricultural Meteorology by OP Bishnoi.

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| <b>AGR0N-413</b> | <b>Climate Resilient Agriculture**</b> | <b>4(3+1)</b> |
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### Objectives

1. To impart the concept of climate resilient agriculture under the present context of climate change
2. To study the integrated role of different sectors in building resilience to climate change in agriculture

## Theory

Climate change and impacts of climate change on agriculture and food security; crop productivity under different climate change scenarios including extreme events such as drought, flood, pest and disease outbreak etc. Basics of adaption and mitigation in the agricultural sectors; analyzing and assessing climate vulnerability to identify vulnerable sectors and possible adaptation options in agriculture; assessing biophysical and socio-economic impacts on agricultural sector; risk assessment strategies, preparedness for weather and climate risks in agriculture; application of geospatial tools and techniques for sustainable agriculture. Climate resilient agriculture (CRA) – concept, scope and importance with special reference to India, climate resilient technologies for enhancing crop productivity and sustainability – role of weather and climatic information, agroadvisories, ICTs and simulation models; climate resilient agronomic practices – crop/cultivar selection, crop diversification/ crop mixtures; water management practices – rain water harvesting, micro-irrigation, deficit irrigation and drainage management, organic/natural farming, integrated farming systems (IFS); site specific nutrient management (SSNM), conservation agriculture technologies to build soil organic carbon, harnessing microbial biodiversity, biomass recycling; use of renewable sources of energy; climate resilient pest-disease management strategies. Breeding strategies for development of climate change resilient crops and varieties, development of biotic and abiotic stress tolerant/resistant cultivars under changed climatic scenarios including extreme weather events.

## Practical

Acquaintance with meteorological instruments including AWS, Statistical techniques to study trend of climatic parameters, Analysis of extreme weather events using non-parametric tests, Building climate change scenarios under different futuristic emission of GHGs, Designing strategies to mitigate the effect of climate change using climate resilient crops/cultivars, Climate resilient technologies and manipulation of cropping patterns, Acquaintance with ICTs for effective dissemination of local weather information and agro-advisories, Analysing carbon sequestration potential of different agro-ecosystems; Designing climate smart village model considering the availability of resources. Awareness programme on climate change and climate resilient agriculture among farming community.

### Lecture schedule: Theory

| S.N. | Topic                                                                         | No. of lectures |
|------|-------------------------------------------------------------------------------|-----------------|
| 1.   | Climate change and impacts of climate change on agriculture and food security | 2               |

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| 2.  | Crop productivity under different climate change scenarios including extreme events such as drought, flood, pest and disease outbreak etc. | 2 |
| 3.  | Basics of adaption and mitigation in the agricultural sectors                                                                              | 1 |
| 4.  | Analyzing and assessing climate vulnerability to identify vulnerable sectors and possible adaptation options in agriculture                | 2 |
| 5.  | Assessing biophysical and socio-economic impacts on agricultural sector                                                                    | 1 |
| 6.  | Risk assessment strategies, preparedness for weather and climate risks in agriculture                                                      | 2 |
| 7.  | Application of geospatial tools and techniques for sustainable agriculture                                                                 | 1 |
| 8.  | Climate resilient agriculture (CRA) – concept, scope and importance with special reference to India                                        | 2 |
| 9.  | Climate resilient technologies for enhancing crop productivity and sustainability– role of weather and climatic information                | 2 |
| 10. | Agro advisories, ICTs and simulation models                                                                                                | 2 |
| 11. | Climate resilient agronomic practices – crop/cultivar selection, crop diversification/ crop mixtures                                       | 1 |
| 12. | Water management practices – rain water harvesting, micro-irrigation, deficit irrigation and drainage management                           | 2 |
| 13. | Organic/natural farming                                                                                                                    | 1 |
| 14. | Integrated farming systems (IFS)                                                                                                           | 2 |
| 15. | Site specific nutrient management (SSNM)                                                                                                   | 1 |
| 16. | Conservation agriculture technologies to build soil organic carbon, harnessing microbial biodiversity, biomass recycling                   | 2 |
| 17. | Use of renewable sources of energy                                                                                                         | 1 |
| 18. | Climate resilient pest-disease management strategies                                                                                       | 1 |
| 19. | Breeding strategies for development of climate change resilient crops and varieties                                                        | 2 |
| 20  | Development of biotic and abiotic stress tolerant/resistant cultivars under changed climatic scenarios including extreme weather events    | 2 |

**Lecture schedule: Practical**

| S.N. | Topic                                                        | No. of lectures |
|------|--------------------------------------------------------------|-----------------|
| 1.   | Acquaintance with meteorological instruments including AWS   | 2               |
| 2.   | Statistical techniques to study trend of climatic parameters | 2               |

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| 3.  | Analysis of extreme weather events using non-parametric tests                                         | 2 |
| 4.  | Building climate change scenarios under different futuristic emission of GHGs                         | 1 |
| 5.  | Designing strategies to mitigate the effect of climate change using climate resilient crops/cultivars | 1 |
| 6.  | Climate resilient technologies and manipulation of cropping patterns                                  | 1 |
| 7.  | Acquaintance with ICTs for effective dissemination of local weather information and agro-advisories   | 2 |
| 8.  | Analysing carbon sequestration potential of different agro-ecosystems                                 | 1 |
| 9.  | Designing climate smart village model considering the availability of resources                       | 2 |
| 10. | Awareness programme on climate change and climate resilient agriculture among farming community       | 2 |

### Suggested readings

1. Climate Resilient Animal Agriculture by GSLHV Prasada Rao. New India Publishing Agency.
2. Climate Resilient Agriculture Adaptation and Mitigation Strategies by Bhan Manish. New India Publishing Agency
3. Climate-Smart Agriculture Sourcebook. FAO (2013).
4. Implications for Climate Smart Agriculture by Wahid Hasan, Sachin G. Mundhe, Abdul Majid Ansari and Shivani Kumari. Biotech Books, 357p.
5. Climate Resilient Agriculture, Adaptation and Mitigation Strategies by Manish Bhan. New India Publishing Agency, 294p.
6. Climate Change and Agriculture Over India by Prasad Rao. PHI Learning, 352p.
7. Climate Smart Agriculture for Sustaining Crop Productivity and Improving Livelihood Security by Prakash M. Satish Serial Publishing House.178p.
8. Climate Smart Agriculture by Shri Rakesh, Kumar, S. Choudhary, S., Poonam. 2023. Shashwat Publication, Bilaspur, Chhattisgarh – 495001

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|------------------|---------------------------------------------------------|--------|
| <b>AGRON-414</b> | Geo-informatics and Remote Sensing, Precision Farming** | 4(3+1) |
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### Objectives

1. Enabling students acquire knowledge on basics of remote sensing technique for precision farming applications
2. To provide a comprehensive knowledge of remote sensing, precision farming and its benefits in improving crop production and soil health management Theory

### Theory

Introduction and history of remote sensing; sources, Principles of remote sensing, propagation of radiations in atmosphere; Interaction with matter; Application of remote sensing techniques land use soil surveys; crop stress and yield forecasting; Advantages and disadvantages of remote

sensing; Remote sensing institutes in India; Basic Concepts about geoinformatics.

Data sharing; Expert System: Introduction to expert system, Characteristics and features of expert system, Applications of Expert System, Importance of Expert system, Rule based system architecture; Software Agents; Impact of Block chain and it's concepts; Probability and Statistics: Bayes Theorem, correlation and Covariance, Continuous Random variables and probability distribution function, various forms of distributions, central limit theorem; Basics of Machine Learning: Random forest, SVM, ensemble methods; Basics of Deep learning: various model architectures and it's training aspects; Hyperspectral and Thermal Remote Sensing; Proximal Soil and Crop Sensors.

### Practical

Familiarization with different remote sensing equipments and data products, Interpretation of aerial photographs and satellite data for mapping of land resources, Global positioning system (GPS), Basics of Geographic Information System (GIS), Georeferencing of toposheets, Digital soil mapping with different variables, Basics of multivariate data analytics, Principal component analysis and regression applications, clustering methods and geostatistics are essential in agricultural studies.

### Lecture schedule-Theory

| S.N. | Topic                                                                                          | No. of lectures |
|------|------------------------------------------------------------------------------------------------|-----------------|
| 1.   | Introduction and history of remote sensing; sources                                            | 2               |
| 2.   | Principles of remote sensing, propagation of radiations in atmosphere; Interaction with matter | 3               |
| 3.   | Application of remote sensing techniques land use soil surveys                                 | 2               |
| 4.   | Crop stress and yield forecasting                                                              | 2               |
| 5.   | Advantages and disadvantages of remote sensing                                                 | 2               |
| 6.   | Remote sensing institutes in India                                                             | 2               |
| 7.   | Basic Concepts about geoinformatics                                                            | 2               |
| 8.   | Data sharing; Expert System: Introduction to expert system                                     | 2               |
| 9.   | Characteristics and features of expert system                                                  | 2               |
| 10.  | Applications of Expert System, Importance of Expert system                                     | 2               |
| 11.  | Rule based system architecture                                                                 | 2               |
| 12.  | Software Agents                                                                                | 2               |
| 13.  | Impact of Block chain and it's concepts                                                        | 2               |
| 14.  | Probability and Statistics: Bayes Theorem                                                      | 2               |
| 15.  | Correlation and Covariance                                                                     | 2               |

|     |                                                                                |   |
|-----|--------------------------------------------------------------------------------|---|
| 16. | Continuous Random variables and probability distribution function              | 3 |
| 17. | Various forms of distributions                                                 | 2 |
| 18. | Central limit theorem                                                          | 2 |
| 19. | Basics of Machine Learning: Random forest, SVM, ensemble methods               | 3 |
| 20. | Basics of Deep learning: various model architectures and it's training aspects | 3 |
| 21. | Hyperspectral and Thermal Remote Sensing                                       | 2 |
| 22. | Proximal Soil and Crop Sensors                                                 | 2 |

### Lecture schedule: Practical

| S.N. | Topic                                                                                 | No. of lectures |
|------|---------------------------------------------------------------------------------------|-----------------|
| 1.   | Familiarization with different remote sensing equipments and data products            | 2               |
| 2.   | Interpretation of aerial photographs and satellite data for mapping of land resources | 2               |
| 3.   | Global positioning system (GPS)                                                       | 2               |
| 4.   | Basics of Geographic Information System (GIS)                                         | 2               |
| 5.   | Georeferencing of toposheets                                                          | 1               |
| 6.   | Digital soil mapping with different variables                                         | 2               |
| 7.   | Basics of multivariate data analytics                                                 | 1               |
| 8.   | Principal component analysis and regression applications                              | 2               |
| 9.   | Clustering methods and geostatistics are essential in agricultural studies            | 2               |

### Suggested readings

1. Data Analytics in Bioinformatics: A Machine Learning Perspective. Editor (s): Rabinarayan Satpathy, Tanupriya Choudhury, Suneeta Satpathy and Sachi Nandan.
2. Machine Learning Approaches to Bioinformatics by Zheng Rong Yang.
3. Text Book of Remote Sensing and Geographical Information Systems by M. Anji Reddy.
4. Precision Agriculture Technologies for Food Security and Sustainability By A El-Kader, M Sherine, M El-Basioni and M Basma.
5. Principles and Theory of Geoinformatics by P.K. Garg. Khanna Publishers. 296p
6. Advances in Geoinformatics Remote Sensing and GIS by Bhunia, Gouri Sankar, Uday Chatterjee and Gopal Krishna Panda. BIO GREEN
7. Artificial Intelligence: Machine Learning, Deep Learning, and Automation Processes by John Adamssen. Efalon Acies.
8. Remote Sensing and Image Interpretation, 6<sup>th</sup> edn (WSE) Paperback – 1 January 2011, Willey Student Edition.
9. Remote Sensing and Geographic Information by A.M. Chandra and S.K. Ghosh. Narosa.

|                  |                                                                               |               |
|------------------|-------------------------------------------------------------------------------|---------------|
| <b>AGRON-415</b> | <b>Principles and Practices of organic Farming/Conservation Agriculture**</b> | <b>4(3+1)</b> |
|------------------|-------------------------------------------------------------------------------|---------------|

### Objectives

1. To teach students the principles of crop production under organic and conservation agriculture situation
2. To impart practical knowledge of organic and conservation agriculture practices

### Theory

Concept of organic farming, principles and its scope in India; Choice of crops and varieties in organic farming; Nutrient management in organic farming and their sources; Fundamentals of insect, pest, disease and weed management under organic mode of production; Operational structure of NPOP; Certification process and crop standards of organic farming; Processing, labelling, economic considerations and viability, marketing and export potential of organic products. Initiatives taken by Government (central/state), NGOs and other organizations for promotion of organic agriculture. Conservation agriculture: definition, origin, principles, advantages, challenges; Primary practices in conservation agriculture: minimum soil disturbance, crop residue retention, and crop diversification, complementary practices, conservation agriculture vis a vis Climate Smart Agriculture; Organic manures- recommended doses and application in comparison to inorganic fertilizers for major crops. Conservation agriculture with special reference to arid regions.

### Practical

Visit of organic farms to study the various components and their utilization; Preparation of enrich compost, vermicompost and their quality analysis; Method of application of bio-fertilizers; Indigenous technology knowledge (ITK) for nutrient, insect-pest and disease management; Studies in green manuring in-situ and green leaf manuring, Studies on different type of botanicals for insect pest management; Weed management in organic farming; Cost of organic production system; Practices of conservation agriculture.

### Lecture schedule: Theory

| S.N. | Topic                                                                     | No. of lectures |
|------|---------------------------------------------------------------------------|-----------------|
| 1.   | Concept of organic farming, principles and its scope in India             | 3               |
| 2.   | Choice of crops and varieties in organic farming                          | 2               |
| 3.   | Nutrient management in organic farming and their sources                  | 5               |
| 4.   | Fundamentals of insect, pest and disease under organic mode of production | 4               |
|      | Weed management under organic mode of production                          | 3               |

|     |                                                                                                                    |   |
|-----|--------------------------------------------------------------------------------------------------------------------|---|
| 5.  | Operational structure of NPOP                                                                                      | 3 |
| 6.  | Certification process and crop standards of organic farming                                                        | 3 |
| 7.  | Processing, labelling, economic considerations and viability of organic products                                   | 3 |
| 8.  | Marketing and export potential of organic products                                                                 | 3 |
| 9.  | Initiatives taken by Government (central/state), NGOs and other organizations for promotion of organic agriculture | 3 |
| 10. | Conservation agriculture: definition, origin, principles, advantages, challenges                                   | 4 |
| 11. | Primary practices in conservation agriculture:                                                                     | 3 |
| 12. | Minimum soil disturbance, crop residue retention, and crop diversification, complementary practices                | 3 |
| 13. | Conservation agriculture vis a vis Climate Smart Agriculture                                                       | 1 |
| 14. | Organic manures- recommended doses and application in comparison to inorganic fertilizers for major crops          | 3 |
| 15. | Conservation agriculture with special reference to arid regions.                                                   | 2 |

### Lecture schedule: Practical

| S.N. | Topic                                                                                  | No. of lectures |
|------|----------------------------------------------------------------------------------------|-----------------|
| 1.   | Visit of organic farms to study the various components and their utilization           | 2               |
| 2.   | Preparation of enrich compost and their quality analysis                               | 2               |
| 3.   | Preparation of vermicompost and their quality analysis                                 | 2               |
| 4.   | Method of application of bio- fertilizers                                              | 2               |
| 5.   | Indigenous technology knowledge (ITK) for nutrient, insect-pest and disease management | 2               |
| 6.   | Studies in green manuring in-situ and green leaf manuring                              | 1               |
| 7.   | Studies on different type of botanicals for insect pest management                     | 1               |
| 8.   | Weed management in organic farming                                                     | 1               |
| 9.   | Cost of organic production system                                                      | 1               |
| 10.  | Practices of conservation agriculture                                                  | 2               |

### Suggested readings

1. A.C. Gaur. Handbook of Organic farming and biofertilizers.
2. A.K. Dahama. Organic Farming for Sustainable Agriculture. Agrobios (India), Jodhpur.
3. Arun. K. Sharma. Handbook of Organic Farming. Agrobios (India), Jodhpur.



4. S.P. Palaniappan and K. Annadurai. Organic Farming – Theory and Practice. Scientific Publishers. Jodhpur.
5. U. Thapa and P. Tripathy. Organic Farming in India- Problems and Prospects. Agrotech publishing agency, Udaipur.
6. G.K. Veeresh. Organic Farming. Foundation Books. New Delhi.
7. Purshit, S.S. Trends in Organic Farming in India. AgrosBios (India), Jodhpur.
8. Thampan, P.K. Organic Agriculture. Peckay tree Crops Development Foundation, Cochin, Kerala.
9. Sathe, T.V. Vermiculture and Organic Farming. Days Publishing House, New Delhi.
10. Singh, Abhinandan, Pankaj Kumar Ojha and Rahul Kumar, 2018. Conservation Agriculture Technologies. Biotech Books.
11. Acharya Sankar Kr, Sreemoyee Bera, Cornea Saha, Prabhat Kumar, Monirul Haque, Riti Chatterjee and Anwesha Mandal. 2022. Conservation Agriculture Approach and Application. Scholars World. 292p.

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|------------------|--------------------------------|---------------|
| <b>AENGG-411</b> | <b>Protected Cultivation**</b> | <b>4(3+1)</b> |
|------------------|--------------------------------|---------------|

### Objective

To educate students on the scientific and commercial cultivation of important value-added products in protected cultivation.

### Theory

Protected cultivation- importance and scope, status of protected cultivation in India and World, types of protected structure based on site and climate. Cladding material involved in greenhouse/ poly house. Greenhouse design, environment control, artificial lights, Automation. Soil preparation and management, Substrate management. Types of benches and containers, Irrigation and fertigation management. Propagation and production of quality planting material of horticultural crops. Greenhouse cultivation of important horticultural crops-rose, carnation, chrysanthemum, gerbera, orchid, anthurium, liliun, tulip, tomato, bell pepper, cucumber, strawberry, pot plants etc. Cultivation of economically important medicinal and aromatic plants. Off- season production of flowers and vegetables. Insect pest and disease management.

### Practical

Raising of seedlings and saplings under protected conditions, Use of portrays in quality planting material production, Bed preparation and planting of crop for production. Inter cultural operations, Soil EC and pH measurement. Regulation of irrigation and fertilizers through drip, fogging and misting.

| S.N. | Topic<br>3                                                                             | No. of<br>lecture |
|------|----------------------------------------------------------------------------------------|-------------------|
| 1.   | Protective cultivation of horticulture crops –importance and scope in India and World. | 1                 |
| 2.   | Various types of protected cultivation structures based on site and climate.           | 1                 |

|    |                                                                                                       |   |
|----|-------------------------------------------------------------------------------------------------------|---|
| 3. | Introduction to green house technology, types of green houses and climate control inside green house. | 2 |
| 4  | Low tunnels for protected cultivation .                                                               | 1 |
| 5  | Shednet house for protected cultivation .                                                             | 1 |
| 6  | Cladding materials involved in greenhouse/ poly house.                                                | 1 |
| 7  | Planning and design of greenhouses.                                                                   | 1 |
| 8  | Materials of construction for traditional and low cost green houses.                                  | 1 |
| 9  | Irrigation systems used in protected cultivation structures.                                          | 1 |
| 10 | Soil preparation and soil management for protected cultivation.                                       | 1 |
| 11 | Types of benches and containers in protected cultivation structures.                                  | 1 |
| 12 | Fertigation management in protected cultivation structures.                                           | 1 |
| 13 | Automation in protected cultivation.                                                                  | 1 |
| 14 | Various equipments and instruments use in protected cultivation.                                      | 1 |
| 15 | Concept of vertical farming inside protected cultivation structures.                                  | 1 |
| 16 | Propagation and production of quality planting material of horticultural                              | 1 |
| 17 | Greenhouse cultivation of Rose and Carnation.                                                         | 2 |
| 18 | Greenhouse cultivation of Chrysanthemum and Gerbera.                                                  | 2 |
| 19 | Greenhouse cultivation of Tomato and Capsicum.                                                        | 2 |
| 20 | Greenhouse cultivation of Cucumber.                                                                   | 1 |
| 21 | Greenhouse cultivation of Strawberry.                                                                 | 1 |
| 22 | Cultivation of economically important medicinal and aromatic plants                                   | 2 |
| 23 | Off- season production of flowers and vegetables in protected cultivation                             | 1 |
| 24 | Disease management in protected cultivation.                                                          | 1 |
| 25 | Insect pest management in protected cultivation.                                                      | 1 |
| 26 | Repair and maintenance of greenhouses.                                                                | 1 |
| 27 | Use of green house in drying                                                                          | 1 |

### Lecture schedule: Practical

| S.No. | Topic                                                                                                | No. of lecture |
|-------|------------------------------------------------------------------------------------------------------|----------------|
| 1.    | Study of various shapes of green houses.                                                             | 1              |
| 2.    | Measurement of climatic factors inside and outside green houses and study of green house equipments. | 1              |
| 3.    | Construct of low tunnels in vegetable crops.                                                         | 1              |
| 4.    | Study of Shade net house and visit to nearby shade net house.                                        | 1              |
| 5     | Drying of agriculture produce in green house.                                                        | 1              |
| 6     | Raising of seedlings in portrays .                                                                   | 1              |
| 7     | Study of irrigation systems in green houses.                                                         | 1              |
| 8     | Plastic mulch laying in greenhouses and low tunnels.                                                 | 1              |
| 9     | Bed preparation in green houses .                                                                    | 1              |
| 10    | Soil and water analysis for protected cultivation (EC, pH).                                          | 1              |
| 11    | Estimation of water requirement under protected cultivation.                                         | 1              |
| 12    | Fertilizer application through drip in protected structures.                                         | 1              |
| 13    | Laying of foggers and misters in protected structures.                                               | 1              |
| 14    | Cultivation of off season vegetables in low tunnels.                                                 | 1              |

|    |                                                      |   |
|----|------------------------------------------------------|---|
| 15 | Visits of nearby poly houses and low tunnel fields . | 2 |
|----|------------------------------------------------------|---|

### Suggested readings

1. Greenhouse operation and management by Paul V. Nelson.
2. Protected cultivation of Horticultural crops by Madan Kr. Jha, Sujana Singh Paikra and Manju Rani Sahu.
3. Protected Cultivation of Horticulture Crops by Itigi Prabhakar. IBPSS.
4. Advances in Protected Cultivation by Brahma Singh and Balraj Singh. NIPA, 252p.
5. Protected Cultivation and Smart Agriculture by Eds. Sagar Maitra, Dinkar J. Gaikwad and Tanmoy Shankar. New Delhi Publishers, 263p.
6. Text book of Protected Cultivation and Precision Farming for Horticultural Crops by B. Ashok Kumar, Eggadi Ramesh and Sindhu V. Jain Brothers.

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|------------------|----------------------------------|---------------|
| <b>AGEXT-411</b> | <b>Agricultural Journalism**</b> | <b>4(3+1)</b> |
|------------------|----------------------------------|---------------|

### Objectives

To impart knowledge and skill in agricultural journalism

### Theory

Journalism – Meaning, nature, importance, and types of journalism. Agricultural Journalism – Meaning, definition, principle, objectives, types, and scope. Similarities and difference between agricultural journalism and other types of journalism. Role of agricultural journalist, Training of agricultural journalist. Qualities of journalist, Role of journalist /journalism in agricultural development and development of newspaper and magazines readers. Newspaper and magazines as communication media: Characteristics, kinds and functions of newspaper and magazines, Characteristics of newspaper and magazines readers. Form, content, style and language of newspaper and magazines, Standard part of newspaper and magazines. The agricultural story: Types of Agriculture stories, subject matter of the agricultural story, structure of the agricultural story. Gathering farm information -Sources of farm information: abstracting from research and scientific materials, interviews, coverage of events. Other sources: electronic media, field study. Success stories definition, nature, components, guidelines of writing a success story. Writing a news story difference between news and feature story, the principle of writing a news story, Inverted pyramid structure. Organizing the material, treatment of the story, writing the news lead and the body. Readability measure-readability ease score, automated readability index, gunning fog index, How to improve readability of articles and stories. Use of photograph in agricultural journalism- Basic principles of photography – composition, exposure, lens, light. Use of artwork (Graphs, charts maps, etc.). Writing the captions. Editorial mechanism: Copy reading, headline and title writing. Proofreading: definition, signs and symbols of proofreading, level of proofreading, duties of a proof-reader. Layout – meaning, principles of layout and design.

**Practical**

Practice in writing an agricultural news story. Practice in writing an agricultural feature story. Covering agricultural events for the information collection. Practice in interviewing for the information collection. Abstracting stories from research and scientific materials and wire services. Selecting pictures and artwork for the agricultural story. Practice in editing, copy reading. Practice in headline and title writing. Practising proof reading. Practice in lay outting of newspaper. Testing copy with a readability formula. Visit a publishing office.

**Suggested readings**

1. Introduction to Journalism by Carole Fleming, Emma Hemmingway, and Gillian Moore.
2. Basic Journalism by Rangaswami Parthasarathy.
3. News Reporting and Editing by K. M. Shrivastava.
4. Professional Journalism by M.V. Kamath.
5. The Journalist's Handbook Book by M.V. Kamath.
6. Farm Journalism and Media Management – Bhaskaran et al.
7. Agricultural Extension and farm Journalism – A K Singh.
8. Farm Journalism – Jana and Mitra.
9. Web Materials.
10. Prepared You Tube videos.

|               |                                            |               |
|---------------|--------------------------------------------|---------------|
| <b>BT-411</b> | <b>Biotechnology of Crop Improvement**</b> | <b>4(3+1)</b> |
|---------------|--------------------------------------------|---------------|

**Theory:**

Impact of Biotechnology on crop improvement and the perspective of society; Various biotechnological techniques available for crop improvement - Plant Tissue Culture, Genetic Engineering, Genome editing, Marker Assisted breeding and Genomic Selection. Biosafety regulations and their application in Agricultural Biotechnology. Somaclonal variation and its use in crop improvement; embryo culture; anther/pollen culture; somatic embryogenesis; artificial seeds; techniques of protoplast culture, regeneration and somatic cell hybridization, achievements and limitations, utility in the improvement of crop plants. Direct and Indirect methods of gene transfer in plants – *Agrobacterium* - mediated gene transfer in dicots and monocots; Direct DNA delivery methods (microinjection, particle gun method, electroporation); gene targeting; Gene silencing techniques; introduction to siRNA; siRNA technology; Micro RNA; construction of siRNA vectors; principle and application of gene silencing; creation of transgenic plants; debate over GM crops; introduction to methods of genetic manipulation in different model systems. Introduction to genome editing - Various tools of genome editing; CRISPR-Cas9 with specific emphasis on Indian regulations; Cloning genomic targets into

CRISPR/Cas9 plasmids; electroporation of Cas9 plasmids into cells; purification of DNA from Cas9 treated cells and evaluation of Cas9 gene editing; *in vitro* synthesis of single guide RNA (sgRNA); using Cas9/sgRNA complexes to test for activity on DNA substrates; evaluate Cas9 activity by T7E1 assays and DNA sequence analysis; Applications of CRISPR/cas9 technology in crop plants. Marker Assisted Breeding & Genomic Selection: Introduction to various DNA-based markers and their use in marker-assisted breeding; Foreground Selection, Recombinant Selection and background Selection; Marker-assisted backcross breeding, marker-assisted selection success stories; Introduction to Genomic Selection.

### Practical:

*Agrobacterium*-mediated transformation in Tobacco, preparation of construct, transfer to binary vector, transform *Agrobacterium*, prepare explant, Inoculation and Co-cultivation, antibiotic based selection of putative transformants, validation using PCR; Genome editing- preparation of CRISPR/Cas construct, direct transfer to plant, analysis of the targets; Planning of a MABB programme selection of parents, crossing strategies, marker analysis

### Theory Schedule

| Unit           | Topic                                                                                                                                                         | Lectures |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1              | Introduction to Agricultural Biotechnology- Impact on crop improvement; societal implications                                                                 | 4        |
| 2              | Overview of Biotechnology Tools - Plant tissue culture, genetic engineering, genome editing, marker-assisted breeding, genomic selection                      | 5        |
| 3              | Biosafety regulations and GM debate Indian biosafety regulations; concerns over GM crops                                                                      | 3        |
| 4              | Somaclonal variation and <i>in vitro</i> techniques: Somaclonal variation, embryo culture, anther/pollen culture, somatic embryogenesis, artificial seeds     | 3        |
| 5              | Protoplast culture and somatic hybridization techniques, achievements, limitations, utility                                                                   | 4        |
| 6              | Gene transfer methods in plants (Direct & Indirect) <i>Agrobacterium</i> -mediated transfer in dicots and monocots; microinjection, gene gun, electroporation | 6        |
| 7              | Gene silencing and regulatory RNAs siRNA, miRNA, vector construction, gene silencing applications                                                             | 4        |
| 8              | Genome editing technologies ZFN, TALENs, CRISPR-Cas9 overview                                                                                                 | 4        |
| 9              | CRISPR-Cas9: methodology and applications designing sgRNA, Cas9 vectors, DNA cleavage assays, application in crop plants                                      | 6        |
| 10             | Marker-Assisted Breeding (MAB); Types of markers, selection types, marker-assisted backcross breeding                                                         | 5        |
| 11             | Genomic selection (GS), introduction, concepts, case studies                                                                                                  | 4        |
| Total lectures |                                                                                                                                                               | 48       |

**Practical Schedule**

| Class | Topic                                                        |
|-------|--------------------------------------------------------------|
| 1     | Preparation of binary vector and Agrobacterium construct     |
| 2     | Agrobacterium transformation and culture preparation         |
| 3     | Explant preparation and co-cultivation techniques            |
| 4     | Antibiotic-based selection and regeneration of transformants |
| 5     | PCR validation of putative transformants                     |
| 6     | Design of CRISPR/Cas construct and in-silico sgRNA design    |
| 7     | In vitro synthesis of sgRNA and preparation of Cas9 complex  |
| 8     | Direct gene transfer into plants using CRISPR construct      |
| 9     | Genomic DNA isolation and PCR from edited plants             |
| 10    | Evaluation of gene editing via T7E1 assay                    |
| 11    | Sequence analysis of edited regions                          |
| 12    | Design of a MABB program: Selection of parents and markers   |
| 13    | Foreground, recombinant, and background marker analysis      |
| 14    | Case study analysis of marker-assisted selection in crops    |
| 15    | Biosafety form design and regulatory checklist               |
| 16    | Final viva, record submission, and project presentation      |

**Suggested readings:**

1. Old, R. W., Primrose, S. B., & Twyman, R. M. (2001). *Principles of Gene Manipulation and Genomics*, 7th Ed. Blackwell.
2. Green, M. R., & Sambrook, J. (2012). *Molecular Cloning: a Laboratory Manual*. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
3. Brown, T. A. (2006). *Genomes* (3rd ed.). New York: Garland Science Pub.
4. Sander JD and Joung JK. (2014). CRISPR-Cas systems for Editing, Regulating and Targeting Genomes. *Nat. Biotechnol.* 32:347-355.
5. *Gene Cloning and DNA Analysis*, 2010
6. Pranav Kumar and Usha Mina, 2015. *Biotechnology: A Problem Approach*, Pathfinder Publication,
7. K. H. Singh, Ajay Kumar and Nehanjali Parmar, 2019. *Agricultural Biotechnology: At a Glance*, science technology.
8. Hari Har Ram, 2019. *Crop Breeding and Biotechnology*: Kalyani Publications
9. S.C. Rastogi, 2020. *Biotechnology: Principles and Applications*, Narosa.
10. Slater, 2008. *Plant Biotechnology: The Genetic Manipulation of Plants*: Oxford, 400p.

|                 |                                    |        |
|-----------------|------------------------------------|--------|
| <b>ENTO-411</b> | Bio-pesticides & Bio-fertilizers** | 4(3+1) |
|-----------------|------------------------------------|--------|

### Objectives

1. To provide knowledge on principles, methods, and mechanisms of bio-control agents and their use against plant diseases
2. To provide knowledge on principles, methods, and mechanism of biofertilizers and their use in agriculture

### Theory

History and concept of bio pesticides. Importance, scope and potential of bio pesticides.

Definitions, concepts and classification of bio pesticides viz. Pathogen, botanical pesticides, and bio rationales. Botanicals and their uses. Mass production technology of bio-pesticides. Virulence, pathogenicity and symptoms of entomopathogenic pathogens and nematodes, Methods of application of bio pesticides. Methods of quality control and Techniques of bio pesticides. Impediments and limitation in production and use of bio pesticides.

Biofertilizers - Introduction, status and scope. Structure and characteristics features of bacterial biofertilizers – Azospirillum, Azotobacter, Bacillus, Pseudomonas, Rhizobium and Frankia; Cyanobacterial bio fertilizers- Anabaena, Nostoc, Hapalosiphon and fungal biofertilizers AM mycorrhiza and ectomycorrhiza. Nitrogen fixation –Free living and symbiotic nitrogen fixation. Mechanism of phosphate solubilisation and phosphate mobilization, K solubilisation. Production Technology: Strain selection, sterilization, growth and fermentation, mass production of carrier based and liquid biofertilizers. FCO specifications and quality control of biofertilizers. Application technology for seeds, seedlings, tubers, sets etc. Biofertilizers-Storage, shelf life, quality control and marketing. Factors influencing the efficiency of biofertilizers.

### Practical

Isolation and purification of important biopesticides: trichoderma Pseudomonas, Bacillus, Metarhizium etc. and its production. Identification of important botanicals. Visit to biopesticide laboratory in nearby area. Field visit to explore naturally infected cadavers. Identification of entomopathogenic entities in field condition. Quality control of biopesticides. Isolation and purification of Azospirillum, Azotobacter, Rhizobium, P-solubilizers and cyanobacteria. Mass multiplication and inoculum production of biofertilizers. Isolation of AM fungi- Wet sieving method and sucrose gradient method. Mass production of AM inoculants.

### Suggested readings

1. Baker, E.F. and James, R.C. 1982. Biological Control of Plant Pathogens. American Phytopathological Society
2. Bhatnagar, R.K. and Palta, R.K. Earthworm Vermiculture and Vermicomposting. Kalyani Publishers.
3. Boland, G.J. and David, L. 1998. Plant Microbe Interactions and Biological Control. Kuykendall Marel Dekker, INC.

4. Borkar, S.G. 2015. Beneficial Microbes as Biofertilizers and its Production Technology.
5. Ciancia, A. and Mukerji, K.J. 2007. General Concepts of Integrated Pest and Disease Management. Edited Published by Springer.
6. Cincholkar, S.B. and Mukherji, K.G. 2007. Biological Control of Plant Diseases. Hawarth Food and Agricultural Products.
7. Gehlot, Dushyent. Organic Farming: Standards, Accreditation, Certification and Inspection. Agrobios (India).
8. Gnanamanickam, S.S. 2002. Biological Control of Crop Disease. Kuykendall Marel Dekker, INC.
9. Nehra, Sampat. Biofertilizers for Sustainable Agriculture. Aavishkar Publishers, Jaipur, India.
10. Ramanujam, B. and Rabindra, R.J. 2006. Current Status of Biological Control of Plant Disease using Antagonistic Organisms in India. Precision Fototype Services, Bengaluru.
11. Singh, Awani Kr. Handbook of Microbial Biofertilizers. Agrotech Press, Jaipur, India.
12. Singh, A.K. Organic Farming. New India Publishing Agency, New Delhi.
13. Singh, S.P. and Hussanini, S.S. 1998. Biological Suppression of Plant Disease, Phytoparasitic Nematodes and Weeds. Precision Fototype Services, Bengaluru.
14. Trivedi, P.C. Fungal Biopesticides and VAM applications. Pointer Publishers, Jaipur, India.

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|-----------------|-----------------------------------------------|---------------|
| <b>ENTO-412</b> | <b>Bio-formulation and Nano Formulation**</b> | <b>4(3+1)</b> |
|-----------------|-----------------------------------------------|---------------|

### **Objectives**

1. To enable students to acquire expertise and skill to develop bioformulation and Nanoformulation
2. To know the importance of biopesticides and biofertilizers
3. To make the students know about various techniques involved in biofertilizers and biopesticides production
4. To get knowledge on essential oils, botanicals, predators, parasitoids, pheromones, and parapheromone and their application in insect pest management
5. To get concepts on agrochemical formulations with nanoparticles and acquaint them with nanotechnology.

### **Theory**

Introduction and history of biological control of pests and diseases; Microbial biopesticides: the global and Indian market scenario; biopesticides for organic agriculture; Different phytopathogenic biocontrol agents: Mode of action; Different entomopathogenic biocontrol agents: Mode of action; Microbial inoculants as biofertilizer candidates, Production, quality assessment and methods of application of biopesticides and biofertilizers; Regulatory system of biopesticides in India; Formulations of plant essential oils, botanicals, pheromone, and parapheromone and their application in insect pest management; Use of predators and parasitoids for insect pest management;

Nanotechnology: its applications in pest and disease diagnosis and management; Nano biopesticides: Concept and importance, different techniques of producing nano biopesticides; Nano Fertilizers: Concept and importance, Types of nano fertilizers; Different techniques of



producing nano fertilizers; Green synthesis of nano fertilizers; green slow-release fertilizer composition based on urea-modified hydroxyapatite nanoparticles

### Practical

Introduction and acquaintance with biopesticide laboratory; Preparation of culture media; Isolation and purification of bioagent from soil and infected insects; Microscopic study of different microbial bioagents; In vitro assay of microbial bioagents against plant pathogens.

In vitro compatibility study among different microbial bioagents; Mass multiplication of biopesticides; Population enumeration of biocontrol agents in different biopesticides; Preparation of plant extracts and their efficacy test against insect pests; Use of pheromone parapheromone for monitoring and management of insect pests; Bioassay of Entomopathogenic biocontrol agents on insect pests; Preparation of microbial inoculants of biofertilizer microbes; Compatibility of biofertilizer microbes; Preparation of solid and liquid consortia of biofertilizer microbes

### Suggested readings

1. Baker, E.F. and James, R.C. 1982. Biological Control of Plant Pathogens. American Phytopathological Society.
2. Borkar, S.G. 2015. Beneficial Microbes as Biofertilizers and its Production Technology.
3. Boland, G.J. and David, L. 1998. Plant microbe interactions and Biological Control. Kuykendall Marel Dekker, INC.
4. Ciancia, A. and Mukerji, K.J. 2007. General Concepts of Integrated Pest and Disease Management. Edited Published by Springer.
5. Cincholkar, S.B. and Mukherji, K.G. 2007. Biological Control of Plant Diseases. Hawarth Food and Agricultural products.
6. Gnanamanickam, S.S. 2002. Biological Control of Crop Disease. Kuykendall Marel Dekker, INC.
7. Ramanujam, B. and Rabindra, R.J. 2006. Current Status of Biological Control of Plant Disease using Antagonistic Organisms in India. Precision Fototype Services, Bengaluru.
8. Singh, S.P. and Hussanini, S.S. 1998. Biological Suppression of Plant Disease, Phytoparasitic Nematodes and Weeds. Precision Fototype Services, Bengaluru.
9. Allhoff, Fritz and Lin, Patrick (Eds). 2009. Nanotechnology and Society. ISBN: 978-1-4020-6208-7 Springer Publications, UK.
10. Prasad, Ram, Vivek Kumar, Manoj Kumar and Devendra Choudhary Eds, 2019. Nanobiotechnology in Bioformulations, Kindle Edition
11. Koul, Opendar Ed, 2019. Nano-biopesticides Today and Future Perspectives. Academic Press.
12. Shah, M. A. and Tokeer Ahmad. Nano Science and Technology, Wiley India.

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|----------------|-----------------------------|--------|
| <b>FSN-411</b> | Food Safety and Standards** | 4(3+1) |
|----------------|-----------------------------|--------|

### Objectives:

1. To develop the skills to convert raw materials in to safe, attractive food products
2. To manage the production of food products
3. To use scientific knowledge to develop new products

### Theory

Food safety – Definition, Importance, Scope and Factors affecting Food Safety. Hazards and Risks, Type of Hazards - Biological, Chemical Physical hazards. Management of hazards – Need. Control of Parameters. Temperature Control. Food Storage. Production Design. Hygiene and Sanitation in Food Service Establishments-Introduction. Sources of contamination and their control. Waste Disposal. Pest and Rodent Control .Personnel Hygiene .Food safety Measures. Food Safety Management Tool- Basic concepts. PRPs, GHPs, GMPs, SSOPs etc. HACCP.ISO series. TQM- concept and need for quality ,components of TQM, Kaizen. Risk Analysis. Accreditation and Auditing, Water Analysis, Surface Sanitation and Personal Hygiene. Food laws and Standards Indian Food Regulatory Regime, FSSAI. Global Scenario CAC. Other laws and standards related to food. Recent concerns-New and Emerging Pathogens. Packaging, Product labelling and Nutritional labelling. Genetically modified food/transgenic. Organic foods. Newer approaches to food safety. Recent Out breaks. Indian and International Standards for food products.

### Practical

Water quality analysis physico-chemical and microbiological. Preparation of different types of media. Microbiological examination of different food samples. Assessment of surface sanitation by swab/rinse method. Assessment of personal hygiene. Biochemical tests for identification of bacteria. Scheme for the detection of food borne pathogens. Preparation of plans for Implementation of FSMS-HACCP, ISO: 22000.

### Suggested readings:

1. Text book of Food Science and Technology by Avantina Sharma
2. Handbook of Food Safety: D. S. L. Khatekar and N.Sarkate, 2023, Step Up Academy, 576p.
3. Food And Beverage Management: Bernard Davis, Andrew Lockwood, Ioannis Pantelidis, Peter Alcott, 2011, Rout ledge
4. Food safety and Quality Control: Pulkit Mathur, 2018, The Orient Blackswan.332p.
5. Safe Food Handling: HACCP booklet for food handlers, Cletus Fernandes, Notion Press

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|---------|----------------------------|--------|
| FSN-412 | Food Science & Nutrition** | 4(3+1) |
|---------|----------------------------|--------|

**Objectives:** To impart knowledge on the biochemical aspects of various nutrients and their interactions in foods during processing, storage and deterioration

### Theory:

Introduction on fundamental aspects of foods and human nutrition; Basic food groups; Concept of balanced diets; Recommended Daily Allowances (RDA) for various age groups; Biochemical composition, energy and food value of various food grains, fruits and vegetables; Carbohydrates,

proteins, fats as nutrients and their interactions; Physio-chemical, functional and nutritional characteristics of essential nutrients- sources and functions, Nutritional requirements, malnutrition, inborn errors of metabolism, deficiency diseases; Digestion, absorption, transport and metabolism of nutrients in human system; Protein quality evaluation. Biochemical and nutritional aspects of vitamins, minerals, nutraceuticals, antioxidant, antinutritional factors and biochemistry of post harvest storage, losses during processing. Effect of cooking, processing and preservation on nutrients of different food products, biochemical aspects of food spoilage; Food fads, food safety and quality standards. Enzymes in food industry, food additives, nutritional quality of plant, animal, dairy, marine and fermented products.

### **Practical:**

Proximate analysis of foods; calorific value of foods, Estimation of vitamins, phenols & flavonoids, carotenoids, antinutrients like Phytate/Oxallate, Trypsin and Chymotrypsin inhibitor activities, limiting amino acids in food stuff

### **Suggested readings:**

1. Damodaran S. and Parkin K L(ed.) 2017. Fennema's Food Chemistry. CRC Press
2. GibneyMJ, Lanham-New SA, Cassidy, Aand Voster HH (ed.) 2009. Introduction to Human Nutrition. Wiley-Blackwell
3. Trueman, P.2007. Nutritional Biochemistry. MJP Publishers
4. Rekhi, Tejmeet and HeenaYadav, 2014. Fundamentals of Food and Nutrition by, Elite Publishing House, 257p.
5. Dharmesh Kumar, Food Science and Nutrition by, RANDOM.

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|----------------|------------------------------------|---------------|
| <b>GPB-411</b> | <b>Commercial Plant Breeding**</b> | <b>4(3+1)</b> |
|----------------|------------------------------------|---------------|

### **Objectives**

1. To discuss about hybrid development and various crop improvement aspects of field crops viz., rice, wheat, maize, pearl millet, sorghum, pigeonpea, chickpea, green gram, black gram, lentil, soybean, groundnut, rapeseed-mustard, cotton etc.
2. To provide understanding on tissue culture and biotechnological approaches as alternative strategies for development of line and cultivars
3. To impart knowledge on seed production, release and notification of varieties and PPV&FR Act, 2001

### **Theory**

Types of crops and modes of plant reproduction. Line development and maintenance breeding in self- and cross- pollinated crops (A/B/R and two-line system) for development of hybrids and seed production. Genetic test of commercial hybrids. Advances in hybrid seed production of maize,

rice, sorghum, pearl millet, castor, sunflower, cotton pigeon pea, Brassica etc. Speed Breeding, Breeding Management systems, High-throughput phenotyping and genotyping platforms, Quality seed production of vegetable crops under open and protected environment. Alternative strategies for the development of the line cultivators: haploid inducer, tissue culture techniques and biotechnological tools. IPR issues in commercial plant breeding: DUS testing and registration of varieties under PPV and FR Act. Variety testing, release and notification systems in India. Principles and techniques of seed production, types of seeds, quality testing in self- and cross-pollinated crops.

### Practical

Floral biology in self- and cross- pollinated species, selfing and crossing techniques. Techniques of seed production in self- and cross- pollinated crops using A/B/R and two-line system. Learning techniques in hybrid seed production using male- sterility in field crops. Understanding the difficulties in hybrid seed production. Tools and techniques for optimizing hybrid seed production. Concept of rouging in seed production plot. Concept of line its multiplication and purification in hybrid seed production. Role of pollinators in hybrid seed production. Hybrid seed production techniques in sorghum, pearl millet, maize, rice, rapeseed-mustard, sunflower, castor, pigeon pea, cotton and vegetable crops. Sampling and analytical procedures for purity testing and detection of spurious seed. Seed drying and storage structure in quality seed management. Screening techniques during seed processing, viz. grading and packaging. Visit to public private seed production and processing plants.

### Lecture Schedule: Theory

| S.N. | Topic                                                                                                                                                   | No. of lectures |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1    | Types of crops and modes of plant reproduction                                                                                                          | 2               |
| 2    | Line development and maintenance breeding in self and cross pollinated crops (A/B/R and two line system) for development of hybrids and seed production | 2               |
| 3    | Genetic purity test of commercial hybrids                                                                                                               | 2               |
| 4    | Advances in hybrid seed production of maize, rice, sorghum, pearl millet                                                                                | 2               |
| 5    | Advances in hybrid seed production of castor, sunflower                                                                                                 | 2               |
| 6    | Advances in hybrid seed production of cotton, pigeon pea, brassica                                                                                      | 2               |
| 7    | Quality seed production of vegetable crops under open and protected environment                                                                         | 2               |

|    |                                                                                       |   |
|----|---------------------------------------------------------------------------------------|---|
| 8  | Alternative strategies for the development of the line and cultivars: haploid inducer | 2 |
| 9  | Tissue culture techniques                                                             | 2 |
| 10 | Biotechnological tools                                                                | 2 |
| 11 | IPR issues in commercial plant breeding                                               | 2 |
| 12 | DUS testing and registration of varieties under PPV & FR Act                          | 2 |
| 13 | Variety testing, release and notification systems in India                            | 2 |
| 14 | Principles and techniques of seed production                                          | 2 |
| 15 | Types of seeds                                                                        | 2 |
| 16 | Quality testing in self and cross pollinated crops                                    | 2 |

### Lecture Schedule: Practical

| S.N. | Topic                                                                                            | No. of lectures |
|------|--------------------------------------------------------------------------------------------------|-----------------|
| 1    | Floral biology in self pollinated species                                                        | 1               |
| 2    | Floral biology in cross pollinated species                                                       | 1               |
| 3    | Selfing and crossing techniques                                                                  | 1               |
| 4    | Selfing and crossing techniques                                                                  | 1               |
| 5    | Techniques of seed production in self and cross pollinated crops using A/B/R and two line system | 1               |
| 6    | Learning techniques in hybrid seed production using male-sterility in field crops                | 1               |
| 7    | Understanding the difficulties in hybrid seed production                                         | 1               |
| 8    | Tools and techniques for optimizing hybrid seed production                                       | 1               |
| 9    | Concept of line its multiplication and purification in hybrid seed production                    | 1               |
| 10   | Role of pollinators in hybrid seed production                                                    | 1               |
| 11   | Hybrid seed production techniques in sorghum                                                     | 1               |
| 12   | Hybrid seed production techniques in pearl millet                                                | 1               |
| 13   | Hybrid seed production techniques in maize                                                       | 1               |
| 14   | Hybrid seed production techniques in rice                                                        | 1               |
| 15   | Hybrid seed production techniques in rapeseed-mustard                                            | 1               |
| 16   | Hybrid seed production techniques in sunflower                                                   | 1               |
| 17   | Hybrid seed production techniques in castor                                                      | 1               |
| 18   | Hybrid seed production techniques in pigeon pea                                                  | 1               |
| 19   | Hybrid seed production techniques in cotton                                                      | 1               |

|    |                                                                                      |   |
|----|--------------------------------------------------------------------------------------|---|
| 20 | Hybrid seed production techniques in vegetable crops                                 | 1 |
| 21 | Sampling and analytical procedures for purity testing and detection of spurious seed | 1 |
| 22 | Seed drying                                                                          | 1 |
| 23 | Seed storage structure in quality seed management                                    | 1 |
| 24 | Screening techniques during seed processing viz., grading and packaging              | 1 |
| 25 | Visit to public private seed production units                                        | 4 |
| 26 | Visit to public private seed processing plants                                       | 4 |

### Suggested readings

1. Commercial Plant Breeding at a glance by Phundan Singh, Pratibha Bisen, Reshu Tiwari. DayaPublishing House.
2. Plant Breeding: Principles and Methods by B. D. Singh. Kalyani Publishers.
3. Principles of Plant Breeding (1st & 2nd Edition) by R.W. Allard.
4. Breeding Field Crops by J.M. Poehlman.
5. Commercial Plant Breeding Objective: Phundan Singh, Mridula Billore and Monika Singh. Astral Publishing, 160p.
6. Breeding and Crop Production: H. Padmalatha, Random.
7. Biotechnology for Agricultural Breeding: Mangal, S. K. GeneTech Books

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|----------------|-----------------------------------------|---------------|
| <b>GPB-412</b> | <b>Micro-propagation Technologies**</b> | <b>4(3+1)</b> |
|----------------|-----------------------------------------|---------------|

### Objectives

To educate the students in detail about the sterilization techniques for explants, preparation of stocks and working solution, culturing of explants, regeneration of whole plants from different explants and hardening procedures.

### Theory

Introduction, History, Advantages and limitations. Types of cultures (seed, embryo, organ, callus, cell); Stages of micro propagation; Axillary bud proliferation (Shoot tip and meristem culture, bud culture); Organogenesis (callus and direct organ formation); Somatic embryogenesis; Cell suspension cultures; production of secondary metabolites; Somaclonal variation; Cryopreservation.

### Practical

Identification and use of equipment in tissue culture Laboratory; Nutrition media composition; Sterilization techniques for media, containers and small instruments; Sterilization techniques for

explants; Preparation of stocks and working solution; Preparation of working medium; Culturing of explants: Seeds, shoot tip and single node; Callus induction; Induction of somatic embryos regeneration of whole plants from different explants; Hardening procedures.

#### Lecture Schedule: Theory

| S.N. | Topic                                                                                                   | No. of lectures |
|------|---------------------------------------------------------------------------------------------------------|-----------------|
| 1    | Meaning and concept of <i>in vitro</i> culture and micro-propagation                                    | 2               |
| 2    | Historical milestones of <i>in vitro</i> culture and micro-propagation                                  | 2               |
| 3    | Advancement and future prospects of micro-propagation                                                   | 2               |
| 4    | Totipotency, dedifferentiation                                                                          | 2               |
| 5    | Tissue culture methodology: Sterilization techniques                                                    | 2               |
| 6    | Synthetic and natural media components                                                                  | 2               |
| 7    | Growth regulators used in tissue culture media                                                          | 2               |
| 8    | Environmental requirement                                                                               | 2               |
| 9    | Genetic control of regeneration                                                                         | 2               |
| 10   | Plant regeneration pathways - Organogenesis and Somatic embryogenesis                                   | 2               |
| 11   | Micro-propagation- Definition, methods, stages of micro-propagation and its significance                | 2               |
| 12   | Axillary bud proliferation approach- Shoot tip and meristem culture                                     | 2               |
| 13   | Organogenesis- Purpose, methods and requirements for organogenesis, indirect and direct organogenesis   | 2               |
| 14   | Somatic embryogenesis- Procedures and requirements for organogenesis, indirect and direct embryogenesis | 2               |
| 15   | Differences between somatic and gametic embryogenesis                                                   | 2               |
| 16   | Synthetic seed-Concepts, necessity, procedure and requirements for production of synthetic seeds        | 2               |

#### Lecture Schedule: Practical

| S.N. | Topic                                      | No. of lectures |
|------|--------------------------------------------|-----------------|
| 1    | Laboratory organization                    | 1               |
| 2    | Sterilization techniques for explants      | 2               |
| 3    | Sterilization techniques for glassware     | 2               |
| 4    | Sterilization techniques for plastic wares | 2               |
| 5    | Sterilization techniques for lab wares     | 2               |

|    |                                                                                  |   |
|----|----------------------------------------------------------------------------------|---|
| 6  | Sterilization techniques for working platform                                    | 2 |
| 7  | Preparation of stocks and working solution                                       | 1 |
| 8  | Preparation of stocks and working solution                                       | 1 |
| 9  | Preparation and sterilization of growth regulators                               | 1 |
| 10 | Preparation of working medium                                                    | 2 |
| 11 | Experimentation on determining optimum concentration of growth regulators        | 2 |
| 12 | Callus induction and regeneration of whole plants from different parts of plants | 4 |
| 13 | Direct regeneration into whole plants using bud                                  | 2 |
| 14 | Direct regeneration into whole plants using node                                 | 2 |
| 15 | Direct regeneration into whole plants using other tissues                        | 2 |
| 16 | Induction of somatic embryos                                                     | 1 |
| 17 | Experiments of synthetic seeds production                                        | 1 |
| 18 | Testing storability of synthetic seed                                            | 1 |
| 19 | Germination efficiency of synthetic seed                                         | 1 |

#### Suggested readings

1. Basics of Horticulture by Jitendra Singh
2. Introduction to Horticulture by N. Kumar
3. Handbook of Horticulture by ICAR.
4. Plant Tissue Culture: Basic and Applied by Timir Baran Jha and Biswajit Ghosh. Platinum Publishers. 439p..

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|----------------|-------------------------------------|---------------|
| <b>GPB-413</b> | <b>Commercial Seed Production**</b> | <b>4(3+1)</b> |
|----------------|-------------------------------------|---------------|

#### Objectives

To introduce the basic principles of planting material production at commercial scale and seed quality evaluation

#### Theory

General Principles of Seed Production: Raising the seed crop, Introduction, Procurement of a class of Improved seeds, Reporting to Monitoring or certification Agency, Principles and practices of selection of area and agronomic requirement of seed production of field crops, Importance of isolation distance and Rouging, Principles of hybrid seed production in field crops, Principles and practices of selection of area and agronomic requirement of seed production of horticultural crops, Concept of apomixes, male sterility and self-incompatibility and its application in hybrid seed production of horticultural crops, Farmers participatory seed



production.

General Principles of Seed Processing: Introduction, Objectives of Seed Processing, Seed Drying, Principles of Drying, Water vapour equilibrium, Methods of drying seeds, Cleaning and grading, Air and screen machines, Dimensional separators, Density separators, Surface texture separators, Colour separators, Spiral separators, Electric separators, Vibrator separators, Separation based on Affinity to liquids, Seed treatment, Temperature treatment, Chemical treatment, Bagging and Labelling.

General Principles of Seed Testing: Seed testing-Introduction, Procedure of Seed testing, components of seed quality testing genetic, physical, physiological and seed health testing, Seed sampling, Types of seed sampling, Requirements of sampling, Concept of seed viability and vigour; dormancy, types and principles of seed dormancy, Physiological quality of seed, Principles of seed Germination, types of germination, biochemical and genetic basis.

Seed Certification: History, concept and objectives of seed certification; seed certification agency/organization and staff requirement Indian Minimum Seed Certification Standards (I.M.S.C.S.) - general and specific crop standards including GM varieties, field and seed standards.

Seed Industry and Seed Marketing: Introduction, Evolution of the seed industry, Development of the vegetable and Flower seed industry, Seed marketing – concept, definition and purpose, importance and promotion of quality seed, formal and informal seed supply systems, Seed marketing intelligence and product mix, sales promotion, distribution channels, marketing costs and margins; packaging and labelling, Seed Associations, Factors influencing seed marketing, Seed marketing programs, Seed industry organizations, Marketing of public versus private players, Demand and supply of seed; role of seed replacement rate (SRR), seed multiplication ratio (SMR), economics of seed production; determining seed needs, Seed pricing and price policy, seed processing and / packaging, demand forecasting and factors affecting demand for seeds, effect of price and farm income on seed demand, Role of WTO in seed marketing.

Biotechnology in Seed Technology: History of plant tissue culture, Laboratory organization, Composition of nutrient medium, Micro-propagation, Axillary bud proliferation approach, Meristem and shoot tip culture, Bud culture, Advantages of Micro-propagation, Problems associated with micro-propagation, Synthetic seed production, Types of synthetic seeds, methods of development of synthetic seeds, Components of nutrient media for synthetic seed development, Storage of synthetic seeds, Advantages and limitations of synthetic seed production.

## **Practical**

Planning of Seed Production, requirements for different classes of seeds in field crops - unit area

and rate

Operation and handling of mechanical drying equipment; effect of drying temperature and duration on seed germination and storability seed processing equipment; seed treating equipment.

Seed production in cross pollinated crops with special reference to land, isolation, Planting ratio of male and female lines, synchronization of parental lines and methods to achieve synchrony; supplementary pollination, pollen storage, hand emasculation and pollination in tomato, Hybrid seed production in Maize, detasseling in maize, identification of rogues and pollen shedders, Pollen collection, storage, viability and stigma receptivity; gametocide application and visits to seed production plots etc., Visit to seed processing plant and commercial controlled and uncontrolled Seed Stores, Seed industries and local entrepreneurs visit to nearby areas, Different methods of examination of seeds to assess seed-borne microorganisms and to quantify infection percentage, detection of seed-borne fungi, bacteria and viruses, identification of storage fungi, control of seed-borne diseases, seed treatment methods., Maintenance of aseptic conditions and sterilization techniques, Preparation of nutrient stocks for synthetic media, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Inoculation of explants for micro-propagation, Inoculation of explants for callus induction and subsequently regeneration of plantlets from matured seeds of field and horticultural crops, Synthetic seed preparation.

#### **Lecture Schedule: Theory**

| <b>S.N.</b> | <b>Topic</b>                                                                                                                                                              | <b>No. of lectures</b> |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1           | General Principles of Seed Production: Raising the seed crop, Introduction, Procurement of a class of Improved seeds, Reporting to Monitoring or certification Agency     | 2                      |
| 2           | Principles and practices of selection of area and agronomic requirement of seed production of field crops, Importance of isolation distance and Rouging                   | 2                      |
| 3           | Principles of hybrid seed production in field crops, Principles and practices of selection of area and agronomic requirement of seed production of horticultural crops    | 2                      |
| 4           | Concept of apomixes, male sterility and self-incompatibility and its application in hybrid seed production of horticultural crops, Farmers participatory seed production. | 2                      |

|    |                                                                                                                                                                                                                                                                                                                                      |   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5  | General Principles of Seed Processing: Introduction, Objectives of Seed Processing, Seed Drying, Principles of Drying, Water vapour equilibrium                                                                                                                                                                                      | 2 |
| 6  | Methods of drying seeds, Cleaning and grading, Air and screen machines, Dimensional separators, Density separators, Surface texture separators, Colour separators, Spiral separators, Electric separators, Vibrator separators, Separation based on Affinity to liquids                                                              | 2 |
| 7  | Seed treatment, Temperature treatment, Chemical treatment, Bagging and Labelling.                                                                                                                                                                                                                                                    | 2 |
| 8  | General Principles of Seed Testing: Seed testing-Introduction, Procedure of Seed testing, components of seed quality testing genetic, physical, physiological and seed health testing                                                                                                                                                | 2 |
| 9  | Seed sampling, Types of seed sampling, Requirements of sampling, Concept of seed viability and vigour; dormancy, types and principles of seed dormancy, Physiological quality of seed                                                                                                                                                | 2 |
| 10 | Principles of seed Germination, types of germination, biochemical and genetic basis.                                                                                                                                                                                                                                                 | 2 |
| 11 | Seed Certification: History, concept and objectives of seed certification; seed certification agency/organization and staff requirement                                                                                                                                                                                              | 2 |
| 12 | Indian Minimum Seed Certification Standards (I.M.S.C.S.) - general and specific crop standards including GM varieties, field and seed standards.                                                                                                                                                                                     | 2 |
| 13 | Seed Industry and Seed Marketing: Introduction, Evolution of the seed industry, Development of the vegetable and Flower seed industry                                                                                                                                                                                                | 2 |
| 14 | Seed marketing – concept, definition and purpose, importance and promotion of quality seed, formal and informal seed supply systems, Seed marketing intelligence and product mix, sales promotion, distribution channels                                                                                                             | 2 |
| 15 | marketing costs and margins; packaging and labelling, Seed Associations, Factors influencing seed marketing, Seed marketing programs, Seed industry organizations, Marketing of public versus private players, Demand and supply of seed                                                                                             | 2 |
| 16 | Role of seed replacement rate (SRR), seed multiplication ratio (SMR), economics of seed production; determining seed needs, Seed pricing and price policy, seed processing and / packaging, demand forecasting and factors affecting demand for seeds, effect of price and farm income on seed demand, Role of WTO in seed marketing | 2 |

|    |                                                                                                                                                               |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 17 | Biotechnology in Seed Technology: History of plant tissue culture, Laboratory organization, Composition of nutrient medium                                    | 2 |
| 18 | Axillary bud proliferation approach, Meristem and shoot tip culture, Bud culture, Advantages of Micro-propagation, Problems associated with micro-propagation | 2 |
| 19 | Synthetic seed production, Types of synthetic seeds, methods of development of synthetic seeds                                                                | 2 |
| 20 | Components of nutrient media for synthetic seed development, Storage of synthetic seeds, Advantages and limitations of synthetic seed production.             | 2 |

### Lecture Schedule: Practical

| S.N. | Topic                                                                                                                                                                                            | No. of lectures |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1    | Planning of Seed Production, requirements for different classes of seeds in field crops - unit area and rate                                                                                     | 1               |
| 2    | Operation and handling of mechanical drying equipment; effect of drying temperature and duration on seed germination and storability seed processing equipment; seed treating equipment.         | 1               |
| 3    | Seed production in cross pollinated crops with special reference to land, isolation, Planting ratio of male and female lines, synchronization of parental lines and methods to achieve synchrony | 1               |
| 4    | supplementary pollination, pollen storage, hand emasculation and pollination in tomato                                                                                                           | 1               |
| 5    | Hybrid seed production in Maize, detasseling in maize                                                                                                                                            | 1               |
| 6    | identification of rogues and pollen shedders, Pollen collection, storage, viability and stigma receptivity; gametocide application and visits to seed production plots etc.                      | 1               |
| 7    | Visit to seed processing plant and commercial controlled and uncontrolled Seed Stores                                                                                                            | 1               |
| 8    | Seed industries and local entrepreneurs visit to nearby areas                                                                                                                                    | 1               |
| 9    | Different methods of examination of seeds to assess seed-borne microorganisms and to quantify infection percentage                                                                               | 1               |

|    |                                                                                                                                              |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|---|
| 10 | Detection of seed-borne fungi, bacteria and viruses, identification of storage fungi, control of seed-borne diseases, seed treatment methods | 1 |
| 11 | Maintenance of aseptic conditions and sterilization techniques, Preparation of nutrient stocks for synthetic media                           | 1 |
| 12 | Preparation of MS medium for micro-propagation and Callus induction                                                                          | 1 |
| 13 | Preparation of MS medium for micro-propagation and Callus induction                                                                          | 1 |
| 14 | Inoculation of explants for micro-propagation                                                                                                | 1 |
| 15 | Inoculation of explants for callus induction and subsequently regeneration of plantlets from matured seeds of field and horticultural crops  | 1 |
| 16 | Synthetic seed preparation.                                                                                                                  | 1 |

### Suggested readings

1. Agarwal, R.L. 1997. Seed Technology. 2nd edn. Oxford & IBH.
2. McDonald, M.B. Jr and Copeland, L.O. 1997. Seed Production: Principles and Practices. Chapman & Hall
3. Thompson, J.R. 1979. An Introduction to Seed Technology. Leonard Hill.
4. Singhal, N.C. 2003. Hybrid Seed Production in Field Crops. Kalyani.
5. Justice, O.L. and Bass, L.N. 1978. Principles and Practices of Seed Storage. Castle House Publ.Ltd.
6. Tunwar, N.S. and Singh S.N. 1988. Indian Minimum Seed Certification Standards. CSCB, Ministry of Agriculture, New Delhi.
7. Chawla, H.S. 2008. Introduction to Plant Biotechnology. 2nd edn. Oxford & IBH publishing Co. Ltd. 113-B Shahpur Jat, New Delhi-110049.
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|-----------------|---------------|--------|
| <b>HORT-411</b> | Landscaping** | 4(3+1) |
|-----------------|---------------|--------|

### Objectives

1. To educate the students on designing different styles and types of gardens.
2. To enable the students on Identify different ornamental plants and their utilization in landscaping.
3. To enable the students to design landscapes in software like AUTOCAD, ARCHCADE etc.

### Theory

Importance and scope of landscaping. Principles of landscaping, garden styles and types, terrace gardening, vertical gardening, garden components, adornments, lawn making, rockery, water garden, walk-paths, bridges, other constructed features etc. gardens for special purposes. Trees: selection, propagation, planting schemes, canopy management, shrubs and herbaceous perennials: selection, propagation, planting schemes, architecture. Climber and creepers: importance, selection, propagation,

planting, Annuals: selection, propagation, planting scheme, Other garden plants: palms, ferns, grasses and cacti succulents. Pot plants: selection, arrangement, management. Bio-aesthetic planning: definition, need, planning; landscaping of urban and rural areas, Peri-urban landscaping, Landscaping of schools, public places like bus station, railway station, townships, river banks, hospitals, play grounds, airports, industries, institutions. Bonsai: principles and management, lawn: establishment and maintenance. CAD application.

### Practical

Identification of trees, shrubs, annuals, pot plants; Propagation of trees, shrubs and annuals, care and maintenance of plants, potting and repotting, identification of tools and implements used in landscape design, training and pruning of plants for special effects, lawn establishment and maintenance, layout of formal gardens, informal gardens, special type of gardens (sunken garden, terrace garden, rock garden) and designing of conservatory and lath house. Use of computer software, visit to important gardens/ parks/ institutes.

### Lecture schedule: Theory

| S. No. | Topic                                                                                    | No. of lectures |
|--------|------------------------------------------------------------------------------------------|-----------------|
| 1.     | Importance and scope of landscaping                                                      | 1               |
| 2.     | Principles of landscaping                                                                | 2               |
| 3.     | Garden styles and types                                                                  | 2               |
| 4.     | Terrace gardening                                                                        | 1               |
| 5.     | Vertical gardening                                                                       | 1               |
| 6.     | Garden components                                                                        | 2               |
| 7.     | Garden adornments                                                                        | 1               |
| 8.     | Lawn making                                                                              | 2               |
| 9.     | Rockery                                                                                  | 1               |
| 10.    | Water garden                                                                             | 1               |
| 11.    | Walk-paths, bridges, other constructed features etc.                                     | 2               |
| 12.    | Gardens for special purposes                                                             | 2               |
| 13.    | Trees: selection, propagation, planting schemes, canopy management                       | 2               |
| 14.    | Shrubs and herbaceous perennials: selection, propagation, planting schemes, architecture | 2               |
| 15.    | Climber: importance, selection, propagation, planting                                    | 2               |
| 16.    | Creepers: importance, selection, propagation, planting                                   | 2               |
| 17.    | Annuals: selection, propagation, planting scheme                                         | 2               |
| 18.    | Other garden plants: palms, ferns, grasses, cacti succulents                             | 2               |
| 19.    | Pot plants: selection, arrangement, management                                           | 2               |

|     |                                                                                                    |           |
|-----|----------------------------------------------------------------------------------------------------|-----------|
| 20. | Bio-aesthetic planning: definition, need, planning                                                 | 2         |
| 21. | Landscaping of urban and rural areas                                                               | 2         |
| 22. | Peri-urban landscaping                                                                             | 2         |
| 23. | Landscaping of schools, public places like bus station, railway station                            | 2         |
| 24. | Landscaping of townships, river banks, hospitals, play grounds, airports, industries, institutions | 2         |
| 25. | Bonsai: principles and management                                                                  | 2         |
| 26. | Lawn: establishment and maintenance                                                                | 2         |
| 27. | CAD application                                                                                    | 2         |
|     |                                                                                                    | <b>48</b> |

### Lecture schedule: Practical

| S. No. | Topic                                                                          | No. of lectures |
|--------|--------------------------------------------------------------------------------|-----------------|
| 1.     | Identification of trees, shrubs, annuals, pot plants                           | 2               |
| 2.     | Identification of tools and implements used in landscape design                | 1               |
| 3.     | Propagation of trees, shrubs and annuals                                       | 2               |
| 4.     | Care and maintenance of plants, shrubs and trees                               | 1               |
| 5.     | Potting and repotting                                                          | 1               |
| 6.     | Training and pruning of plants for special effects                             | 1               |
| 7.     | Lawn establishment and maintenance                                             | 1               |
| 8.     | Layout of formal gardens                                                       | 1               |
| 9.     | Layout of informal gardens                                                     | 1               |
| 10.    | Layout of special type of gardens (sunken garden, terrace garden, rock garden) | 2               |
| 11.    | Designing of conservatory and lath house                                       | 1               |
| 12.    | Use of computer software                                                       | 1               |
| 13.    | Visit to important gardens/ parks/ institutes                                  | 1               |
|        |                                                                                | <b>16</b>       |

### Suggested readings

1. Arora, J.S. Introductory Ornamental Horticulture
2. Nambisan, K.M.P. 1992. Design and elements of landscape gardening. Oxford and IBH Publications, New Delhi.
3. Pal, B.P. 1960. Beautiful climbers of India. ICAR, New Delhi
4. Randhava, G.S. 1973. Ornamental Horticulture in India. Today and Tomorrow's Printers and Publishers. Karol Bagh, New Delhi
5. Randhava, G.S and Mukhopadhyay, A. 1986. Floriculture in India. Allied Publishers Pvt. Ltd., New

6. Yadav, P. K. & R. P. Singh 2021. Flower Production & Gardening, New India Publishing Agency, New Delhi, ISBN 9789390591213

|                 |                               |               |
|-----------------|-------------------------------|---------------|
| <b>HORT-412</b> | <b>Hi-tech Horticulture**</b> | <b>4(3+1)</b> |
|-----------------|-------------------------------|---------------|

### Objectives

1. To educate the students on the latest technology of hi-tech horticulture
  2. To educate students on the concepts and prospects of hi-tech horticulture.
1. Students will know about protected cultivation of different high value vegetables like tomato, cucumber and sweet pepper etc.

### Theory

Introduction and importance; Nursery management and mechanization; micro propagation of horticultural crops; Modern field preparation and planting methods; Protected cultivation: advantages, controlled conditions, method and techniques; Micro irrigation systems and its components; EC, pH based fertilizer scheduling; canopy management; high density orcharding; Components of precision farming: Remote sensing; Geographical Information System (GIS); Differential Geo-positioning System (DGPS); Variable Rate Applicator (VRA); application of precision farming in horticultural crops (fruits, vegetables and ornamental crops); mechanized harvesting of produce.

### Practical

Types of polyhouses and shade net houses, Intercultural operations, tools and equipment identification and application, Use of mulching in horticultural crops, Micro propagation, Nursery- portrays, micro-irrigation, EC, pH based fertilizer scheduling, canopy management, visit to hi-tech orchard/nursery.

### Lecture schedule: Theory

| S. No. | Topics                                                      | No. of lectures |
|--------|-------------------------------------------------------------|-----------------|
| 1.     | Introduction of Hi-tech Horticulture                        | 1               |
| 2.     | Importance of Hi-tech Horticulture                          | 2               |
| 3.     | Nursery management                                          | 3               |
| 4.     | Mechanization                                               | 2               |
| 5.     | Micro propagation of horticultural crops                    | 2               |
| 6.     | Modern field preparation                                    | 2               |
| 7.     | Planting methods                                            | 2               |
| 8.     | Protected cultivation: advantages and controlled conditions | 2               |



|     |                                                      |           |
|-----|------------------------------------------------------|-----------|
| 9.  | Protected cultivation: method and techniques         | 3         |
| 10. | Micro irrigation systems and its components          | 2         |
| 11. | EC, pH-based fertilizer scheduling                   | 2         |
| 12. | Canopy management                                    | 3         |
| 13. | High density orcharding                              | 2         |
| 14. | Components of precision farming                      | 3         |
| 15. | Remote sensing                                       | 2         |
| 16. | Geographical Information System (GIS)                | 2         |
| 17. | Differential Geo-positioning System (DGPS)           | 2         |
| 18. | Variable Rate Applicator (VRA)                       | 2         |
| 19. | Application of precision farming in fruits crops     | 2         |
| 20. | Application of precision farming in vegetable crops  | 2         |
| 21. | Application of precision farming in ornamental crops | 2         |
| 22. | Mechanized harvesting of horticultural produce       | 3         |
|     |                                                      | <b>48</b> |

### Lecture schedule: Practical

| S. No. | Topics                                                                        | No. of lectures |
|--------|-------------------------------------------------------------------------------|-----------------|
| 1.     | Types of polyhouses and shade net houses                                      | 2               |
| 2.     | Intercultural operations, tools and equipment identification and application, | 2               |
| 3.     | Use of mulching in horticultural crops                                        | 1               |
| 4.     | Micro propagation,                                                            | 2               |
| 5.     | Nursery- portrays,                                                            | 1               |
| 6.     | Micro-irrigation,                                                             | 2               |
| 7.     | EC, pH based fertilizer scheduling,                                           | 2               |
| 8.     | Canopy management,                                                            | 2               |
| 9.     | Visit to hi-tech orchard                                                      | 1               |
| 10.    | Visit to hi-tech nursery                                                      | 1               |
|        |                                                                               |                 |

### Suggested readings

1. Hi-tech Horticulture by T.A. More.
2. Greenhouse Operation and Management by Paul V. Nelson.
3. Hi Tech Horticulture (Pb) by S. Prasad, Dharam Singh and R.L. Bharadwaj. Agrobios
4. Instant Horticulture by S.N. Gupta. Jain Brothers. 488p.

5. Hydroponics for Beginners and Advanced: The Ultimate Hydroponic and Aquaponic Gardening Guide by Tom Garden, Webb Eleanor.

|                 |                                                     |               |
|-----------------|-----------------------------------------------------|---------------|
| <b>HORT-413</b> | <b>Post-harvest Technology and Value Addition**</b> | <b>4(3+1)</b> |
|-----------------|-----------------------------------------------------|---------------|

### Objectives

1. To educate about the different pre-harvest, harvest and post-harvest factors affecting the postharvest life of fruits and vegetables
2. To educate about preparation techniques of value-added products
3. To educate about the different dehydration techniques of horticultural crops

### Theory

Importance of post-harvest processing of fruits and vegetables, extent and possible causes of post-harvest losses: Pre-harvest factors affecting post-harvest quality, maturity, ripening and changes occurring during ripening; Respiration and factors affecting respiration rate; Harvesting and field handling; Storage (ZECC, cold storage, CA, MA and hypobaric); Value addition concept; Principles and methods of preservation; Intermediate moisture food (jam, jelly, marmalade, preserve, candy) - concepts and standards; Fermented and non-fermented beverages. Tomato products -concepts and standards; Drying /Dehydration of fruits and vegetables –concept and methods, osmotic drying. Canning – concepts and standards, packaging of products.

### Practical

Identification of important tools/equipments/machines and chemicals required for PHT laboratory. Applications of different types of packing, containers for shelf-life extension. Effect of temperature on shelf life and quality of produce. Demonstration of chilling and freezing injury in vegetables and fruits. Extraction and preservation of pulps and juices. Preparation of jam, jelly, RTS, nectar, squash, osmotically dried products, fruit bar candy and tomato products, canned products. Quality evaluation of products-Physico-chemical and sensory. Visit to processing unit/industry.

### Lecture schedule: Theory

| S. No. | Topics                                                         | No. of lectures |
|--------|----------------------------------------------------------------|-----------------|
| 1.     | Importance of post-harvest processing of fruits and vegetables | 1               |
| 2.     | Extent and possible causes of post-harvest losses              | 1               |
| 3.     | Pre-harvest factors affecting post-harvest quality             | 1               |
| 4.     | Maturity indices                                               | 1               |
| 5.     | Ripening and changes occurring during ripening                 | 1               |
| 6.     | Respiration and factors affecting respiration rate             | 1               |

|     |                                                                                              |           |
|-----|----------------------------------------------------------------------------------------------|-----------|
| 7.  | Harvesting and field handling                                                                | 1         |
| 8.  | Storage (ZECC, cold storage, CA, MA and hypobaric)                                           | 1         |
| 9.  | Value addition concept                                                                       | 1         |
| 10. | Principles and methods of preservation                                                       | 2         |
| 11. | Intermediate moisture food (jam, jelly, marmalade, preserve, candy) - concepts and standards | 1         |
| 12. | Fermented and non-fermented beverages                                                        | 1         |
| 13. | Tomato products -concepts and standards                                                      | 1         |
| 14. | Drying /Dehydration of fruits and vegetables –concept and methods, osmotic drying            | 1         |
| 15. | Canning – concepts and standards, packaging of products                                      | 1         |
|     |                                                                                              | <b>16</b> |

#### **Lecture schedule: Practical**

| <b>S. No.</b> | <b>Topics</b>                                                                                   | <b>No. of lectures</b> |
|---------------|-------------------------------------------------------------------------------------------------|------------------------|
| 1.            | Identification of important tools/equipments/machines and chemicals required for PHT laboratory | 1                      |
| 2.            | Applications of different types of packing, containers for shelf-life extension                 | 1                      |
| 3.            | Effect of temperature on shelf life and quality of produce                                      | 1                      |
| 4.            | Demonstration of chilling and freezing injury in vegetables and fruits                          | 2                      |
| 5.            | Extraction and preservation of pulps and juices                                                 | 2                      |
| 6.            | Preparation of jam, jelly, RTS, nectar, squash                                                  | 2                      |
| 7.            | Osmotically dried products                                                                      | 1                      |
| 8.            | Fruit bar candy                                                                                 | 1                      |
| 9.            | Tomato products                                                                                 | 1                      |
| 10.           | Canned products                                                                                 | 1                      |
| 11.           | Quality evaluation of products- Physico-chemical and sensory                                    | 2                      |
| 12.           | Visit to processing unit/industry                                                               | 1                      |
|               |                                                                                                 | <b>16</b>              |

#### **Suggested readings**

1. Post-harvest technology of horticultural crops by S.K. Sharma and M.C Nautiyal.
2. Post-Harvest Technology by Suja Nabi Qureshi, Kounser Javeed and Abhay Kumar Sinha. Bioscientific Publishers.

3. Postharvest Technology of Horticultural Crops by K.P. Sudheer and V. Indira. New India Publishing Agency. 320p.
4. Postharvest Management and Value Addition by Aswini Kumar Goel, Rajender Kumar and Satwinder S. Mann. Daya Publishing House.
5. Postharvest Management and Value Addition of Fruits and Vegetables by Kureel M.K. Biotech, 181p
6. Yadav, P. K. and Singh, J. 2000. Phalotpadan and Prasanskaran Published.in 2000 from Agro- bios (India), Jodhpur., ISBN 81-7754-086-6

|                 |                                 |        |
|-----------------|---------------------------------|--------|
| <b>SSAC-411</b> | Vermicompost and Vermiculture** | 4(3+1) |
|-----------------|---------------------------------|--------|

### Objective

To provide a comprehensive knowledge of vermicomposting, vermiculture and chemical analysis of organic field soils

### Theory

Concept of Vermiculture and vermicompost; Identification of earth worms; Drawing of flow-sheet chart of vermitechnology; Requirements for vermicomposting a. Basic raw material, b. Selection of suitable earthworm, c. Starter, d. Site selection, e. Containers for vermicompost production, f. Thatched roof/vermished; Role of earthworm in vermicomposting a. A brief description of earthworm, b. Suitable earthworm for vermicomposting, c. How does earthworm facilitate vermicomposting?, d. Favourable conditions for earth worm culture in the composting; Methods of vermicomposting a. Bed method- stepwise procedure, b. Pit method- stepwise procedure, c. Re-composition and In-situ vermicomposting; Handling and harvesting of vermicompost, a. Commonly faced problem in vermicomposting, b. Five phases of vermicomposting, c. Precautions to be taken during the five phases, d. Assessing the maturity of vermicompost and harvesting, e. Methods of earthworm harvesting, f. Storing and packing of vermicompost, material; Vermiwash: a. Principle of Vermiwash preparation, b. Method of application of Vermiwash; Benefits of vermicompost: a. Nutrient content of vermicompost, b. Application of vermicompost for different crops, c. Soil health improvement and crop productivity enhancement; Vermicompost : an additional source of income; Potential funding sources for adoption of vermicompost unit at farmers' field

| S. No. | Topic                                          | No. of lecture |
|--------|------------------------------------------------|----------------|
| 1.     | Concept of Vermiculture and vermicompost       | 3              |
| 2.     | Identification of earth worms                  | 3              |
| 3.     | Drawing of flow-sheet chart of vermitechnology | 3              |

|     |                                                                                                                                                                                                                                                                                                                                                                              |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.  | <p>Requirements for vermicomposting</p> <p>a. Basic raw material,</p> <p>b. Selection of suitable earthworm,</p> <p>c. Starter, d. Site selection,</p> <p>e. Containers for vermicompost production,</p> <p>f. Thatched roof/vermished</p>                                                                                                                                   | 6  |
| 5.  | <p>Role of earthworm in vermicomposting</p> <p>a. A brief description of earthworm,</p> <p>b. Suitable earthworm for vermicomposting,</p> <p>c. How does earthworm facilitate vermicomposting?</p> <p>d. Favourable conditions for earth worm culture in the composting</p>                                                                                                  | 6  |
| 6.  | <p>Methods of vermicomposting</p> <p>a. Bed method- stepwise procedure,</p> <p>b. Pit method- stepwise procedure,</p> <p>c. Re-composition and In-situ vermicomposting</p>                                                                                                                                                                                                   | 6  |
| 7.  | <p>Handling and harvesting of vermicompost,</p> <p>a. Commonly faced problem in vermicomposting,</p> <p>b. Five phases of vermicomposting,</p> <p>c. Precautions to be taken during the five phases,</p> <p>d. Assessing the maturity of vermicompost and harvesting,</p> <p>e. Methods of earthworm harvesting,</p> <p>f. Storing and packing of vermicompost, material</p> | 6  |
| 8.  | <p>Benefits of vermicompost:</p> <p>a. Nutrient content of vermicompost,</p> <p>b. Application of vermicompost for different crops,</p> <p>c. Soil health improvement and crop productivity enhancement</p>                                                                                                                                                                  | 5  |
| 9.  | <p>Vermiwash:</p> <p>a. Principle of Vermiwash preparation,</p> <p>b. Method of application of Vermiwash</p>                                                                                                                                                                                                                                                                 | 5  |
| 10. | Vermicompost: an additional source of income                                                                                                                                                                                                                                                                                                                                 | 3  |
| 11. | Potential funding sources for adoption of vermicompost unit at farmers' field                                                                                                                                                                                                                                                                                                | 2  |
|     |                                                                                                                                                                                                                                                                                                                                                                              | 48 |

**Practical**

Determination of N in vermicompost; Determination of phosphorus in vermicompost (volumetric ammonium molybdate method); Determination of potassium in vermicompost; Estimation of organic carbon in vermicompost; Fractionation of vermicompost; Determination of infiltration rate of soil with application of vermicompost; Determination of moisture content of soil with application of vermicompost; Determination of available nitrogen in organic field's soil; Determination of P content of organic field's soil; Determination of K content of organic field's soil; Identification of different strains of bio-fertilizers

| S. No. | Topic                                                                              | No. of lecture |
|--------|------------------------------------------------------------------------------------|----------------|
| 1.     | Determination of N in vermicompost                                                 | 1              |
| 2.     | Determination of phosphorus in vermicompost (volumetric ammonium molybdate method) | 1              |
| 3.     | Determination of potassium in vermicompost                                         | 1              |
| 4.     | Estimation of organic carbon in vermicompost                                       | 1              |
| 5.     | Fractionation of vermicompost                                                      | 2              |
| 6.     | Drawing of flow-sheet chart of vermitechology                                      | 1              |
| 7.     | Determination of infiltration rate of soil with application of vermicompost        | 2              |
| 8.     | Determination of moisture content of soil with application of vermicompost         | 2              |
| 9.     | Determination of available nitrogen in organic field's soil                        | 1              |
| 10.    | Determination of P content of organic field's soil                                 | 1              |
| 11.    | Determination of K content of organic field's soil                                 | 1              |
| 12.    | Identification of different strains of bio-fertilizers                             | 2              |
|        |                                                                                    | 16             |